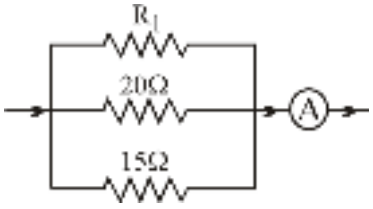
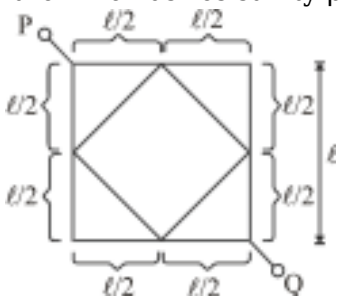
PHYSICS

1) In the given circuit the current flowing through the resistance 20 ohms is 0.3 ampere while the ammetre reads 0.8 ampere. What is the value of R_1?



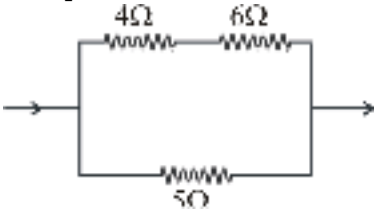
(1) 30 ohms
(2) 40 ohms
(3) 50 ohms
(4) 60 ohms

2) If the wire has resistivity ρ and cross sectional area A, the equivalent resistance between P and Q is :-



(1) $\frac{\rho\ell}{\sqrt{2}A}$
(2) $\frac{\sqrt{2}\rho\ell}{A}$
(3) $\frac{2\rho\ell}{A}$
(4) $\frac{\rho\ell}{A}$

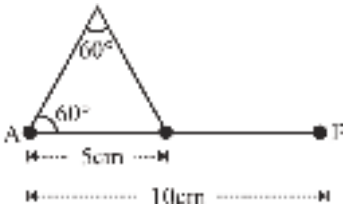
3) In the circuit shown in figure the heat produced in the 5Ω resistor due to the current flowing through it is 10 calories per second. The heat generated in the 4Ω resistor is:



(1) 1 cal/s
(2) 2 cal/s
(3) 3 cal/s

(4) 4 cal/s

4) A wire has resistance of 24 Ω and it is bent in a shape, as shown in figure. The effective resistance between A and B is :-



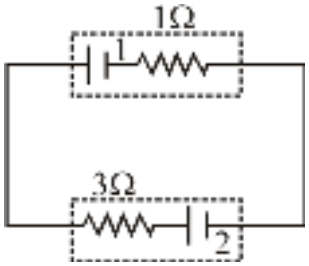
(1) 24 Ω

(2) 10 Ω

(3) $\frac{16}{3}\Omega$

(4) $\frac{18}{5}\Omega$

5) In the figure shown, battery 1 has emf = 6 V and internal resistance = 1 Ω. Battery 2 has emf = 2V and internal resistance = 3 Ω. The wires have negligible resistance. What is the potential difference across the terminals of battery 2 ?



(1) 4 V

(2) 1.5 V

(3) 5 V

(4) 0.5 V

6) An insulating pipe of cross-section area 'A' contains an electrolyte which has two types of ions their charges being –e and +2e. A potential difference applied between the ends of the pipe result in the drifting of the two types of ions, having drift speed = v (–ve ion) and v/4 (+ve ion). Both ions have the same number per unit volume = n. The current flowing through the pipe is :

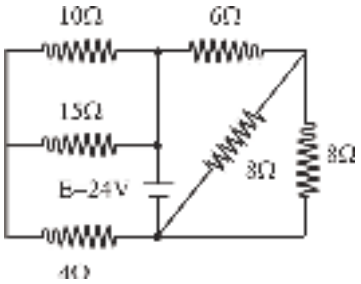
(1) nev A/2

(2) nev A/4

(3) 5nev A/2

(4) 3nev A/2

7) Find the equivalent resistance across the terminals of source of e.m.f. 24 V for the circuit shown



in figure:-

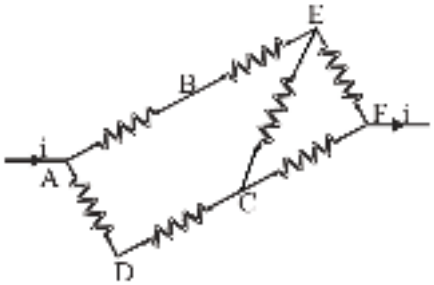
(1) 15 Ω

(2) 10 Ω

(3) 5Ω

(4) 4Ω

8) In the adjoining circuit diagram each resistance is of 10 Ω. The current in the arm AD will be :-



(1) $\frac{i}{2}$

(2) $\frac{3i}{5}$

(3) $\frac{2i}{3}$

(4) $\frac{i}{4}$

9) Two electric bulbs 40W, 200 V and 100 W, 200 V are connected in series. Then the maximum voltage that can be applied across the combination, without fusing either bulb is -

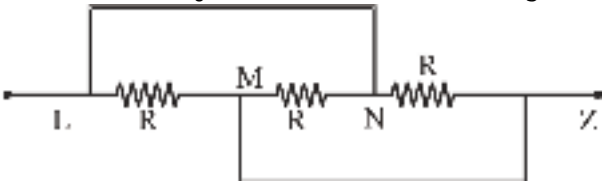
(1) 280 V

(2) 400 V

(3) 3000 V

(4) 200 V

10) Three equal resistances each of value R are joined as shown in the figure. The equivalent resistance between M and N is :-



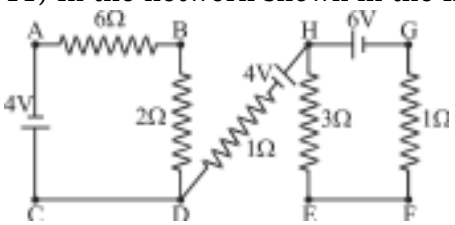
(1) R

(2) 2R

(3) $\frac{R}{2}$

(4) $\frac{R}{3}$

11) In the network shown in the figure, choose the incorrect statement :-

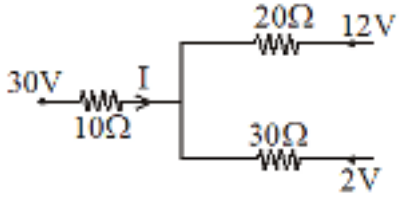


(1) Potential difference between points H and B is 5V

(2) Potential difference between points D and B is 1 V

(3) Potential difference between points D and E is 1 V

(4) Potential difference between points D and G is 2V

12) 

I = ?

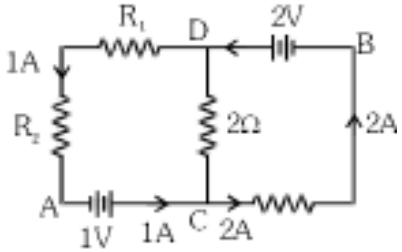
(1) 1A

(2) 2.5A

(3) 4.7A

(4) 10A

13) In the circuit shown in the figure, if the potential at point A is taken to be zero, the potential at point B is :-



(1) +1V

(2) –1V

(3) +2V

(4) –2V

14) You are provided with 48 cells, each of emf 2V and internal resistance 4 Ohms. What maximum current can flow in a circuit having an external resistance of 12Ω.

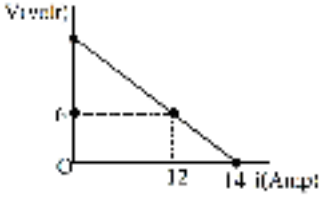
(1) 1A

(2) 1.2 A

(3) 0.96 A

(4) 1.08 A

15) 10 cells, each of emf E and internal resistance are connected in series to a variable external resistance. Figure shows the variation of terminal potential difference of their combination with current drawn from the combination. Emf of each cell is



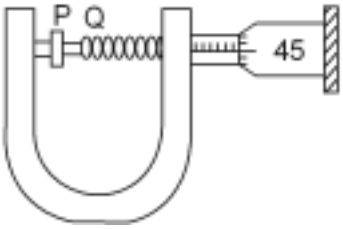
(1) 1.6 V

(2) 3.6 V

(3) 1.4 V

(4) 4.2 V

16) In an experiment to find out the diameter of wire using screw gauge, the following observations were noted :



(a) Screw moves 0.5 mm on main scale in one complete rotation

(b) Total divisions on circular scale = 50

(c) Main scale reading is 2.5 mm

(d) 45th division of circular scale is in the pitch line

(e) Instrument has 0.03 mm negative error

Then the diameter of wire is :

(1) 2.92 mm

(2) 2.54 mm

(3) 2.98 mm

(4) 3.45 mm

17) A particle is projected with $\vec{u} = 30\hat{i} + 40\hat{j}$ m/s from ground. Then match the two columns (in respective units)

	Column-I		**Column-II**
(A)	Time of flight	P.	30
(B)	Maximum height	Q.	80
(C)	Range	R.	240
(D)	Horizontal velocity at t = 3s.	S.	8
		T.	10

(1) A-S, B-Q, C-R, D-P

(2) A-P, B-Q, C-R, D-S

(3) A-S, B-R, C-Q, D-P

(4) A-T, B-Q, C-S, D-P

18) The mass and volume of a body are found to be (5.00 ± 0.05) kg and (1.00 ± 0.05) m^3 respectively. Then the maximum possible percentage error in density :-

(1) 3%

(2) 10%

(3) 6%

(4) 5%

19) The position of an object moving along the x-axis is given by: $x = A + Bt^2$ Where A = 10 m, B = 3 m/s^2, and t is measured in seconds.
(i) What is its velocity at **t = 3.0 s**?
(ii) What is the average velocity between **t = 1.0 s** and **t = 3.0 s**?

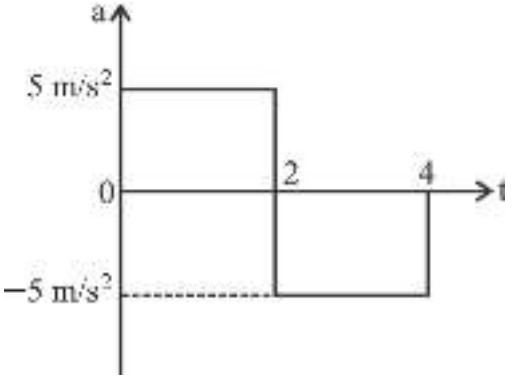
(1) Velocity = 18 m/s, Avg Velocity = 12 m/s

(2) Velocity = 9 m/s, Avg Velocity = 15 m/s

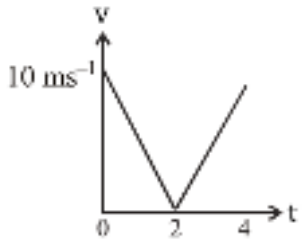
(3) Velocity = 18 m/s, Avg Velocity = 18 m/s

(4) Velocity = 12 m/s, Avg Velocity = 10 m/s

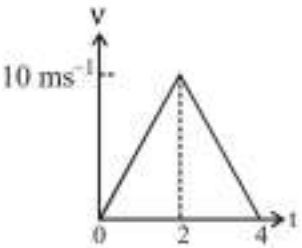
20) A body starts from rest at t = 0 and undergoes an acceleration as shown in the graph. Which velocity time graph represent the motion of the body in 4 second ?

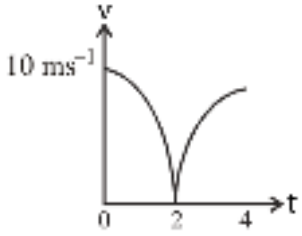


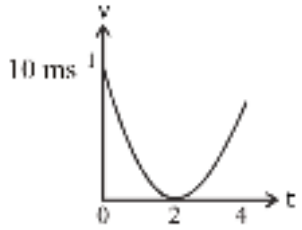
(1)



(2)



(3) 

(4) 

21) A body is moving from rest under constant acceleration and let S_1 be the displacement in the first (p-1) sec and S_2 be the displacement in the first p sec. The displacement in $(p^2 - p + 1)^{th}$ sec. will be : -

(1) $S_1 + S_2$

(2) $S_1 S_2$

(3) $S_1 - S_2$

(4) S_1 / S_2

22) If the time of oscillation (T) of a gas bubble from an explosion under water depends on static pressure (P), density of water (d) and total energy of explosion (E), find the relation between them : -

(1) $T \propto P^{-5/6} d^{1/2} E^{1/3}$

(2) $T \propto P^{1/2} d^{1/3} E^{1/2}$

(3) $T \propto P^{-5/6} d^{1/3} E^{1/2}$

(4) $T \propto P^{-2/3} d^{1/2} E^{1/3}$

23) Water drops fall from a tap located on the roof of a building 16 meters high at regular intervals of time. When the fifth drop is just about to leave the tap, the first drop touches the ground, then the distance of the second drop from the top (the tap) at this instant is. (Take $g = 10\ m/s^2$)

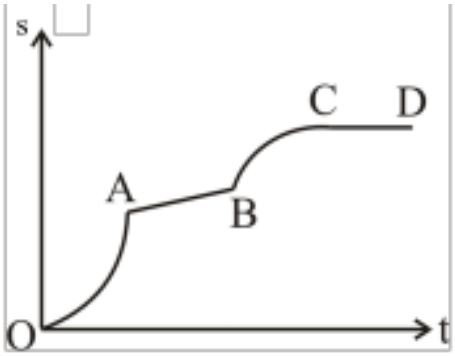
(1) 4 meters

(2) 9 meters

(3) 12 meters

(4) 1 meter

24) The displacement versus time curve is given OA, BC are parabolic and CD is parallel to time axis



	Column-I		Column-II
(i)	OA	(a)	Velocity increases with time
(ii)	AB	(b)	Velocity decreases with time
(iii)	BC	(c)	Velocity independent of time
(iv)	CD	(d)	Velocity is zero

(1) i→b, ii→a, iii→d, iv→c

(2) i→a, ii→c, iii→d, iv→b

(3) i→a, ii→c, iii→b, iv→d

(4) i→d, ii→b, iii→c, iv→d

25) A particle moves along a straight line such that its position at any time t is given by

$S = t^3 - 6t^2 + 3t + 4$

The velocity , when its acceleration is zero is -

(1) 3 ms^{-1}

(2) –12 ms^{-1}

(3) 42 ms^{-1}

(4) –9 ms^{-1}

26) A body moves 6 m north, 8 m east and 10m vertically upwards, what is its resultant displacement from initial position:-

(1) $10\sqrt{2}m$

(2) 10m

(3) $\frac{10}{\sqrt{2}}m$

(4) 10 × 2m

27) A body is moving with velocity 10 m/s towards north and after 2 seconds its velocity becomes 20 m/s towards west. The average acceleration of the body will be :-

(1) $5\sqrt{5}$ m/s^2

(2) 25 m/s^2

(3) 125 m/s^2

(4) 15 m/s^2

28) For motion of an object along the x-axis the velocity v depends on the displacement x as $v = 3x^2 - 2x$, then what is the acceleration at x = 2m.

(1) 48 ms^{-2}

(2) 80 ms^{-2}

(3) 18 ms^{-2}

(4) 10 ms^{-2}

29) Two tall buildings are 30 m apart. The speed with which a ball must be thrown horizontally from a window 150 m above the ground in one building so that it enters a window 27.5 m from the ground in the other building is :-

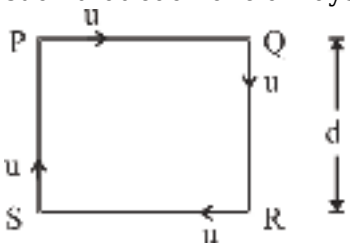
(1) 2 m s^{-1}

(2) 6 m s^{-1}

(3) 4 m s^{-1}

(4) 8 m s^{-1}

30) Four persons P, Q, R and S of same mass travel with same speed u along a square of side 'd' such that each one always faces the other. After what time will they meet each other ?



(1) $\frac{d}{u}$

(2) $\frac{2d}{3u}$

(3) $\frac{2d}{u}$

(4) $d\sqrt{3}u$

31) A train of 150 m length is running towards north at a speed of 10 m/s. A parrot flies at a speed of 5 m/s towards south parallel to the railway track. The time taken by the parrot to cross the train is equal to:–

(1) 12 s

(2) 8 s

(3) 15 s

(4) 10 s

32) A man can swim in still water with a speed 2 times that of the river then at what angle with downstream he must swim to cross the river along shortest possible path :-

(1) 30°

(2) 120°

(3) 150°

(4) 60°

33) An aeroplane moving horizontally with a speed of 720 km/h drops a food pocket, while flying at a height of 396.9 m. The time taken by a food pocket to reach the ground and its horizontal range is (Take g = 9.8 m/sec^2) :-

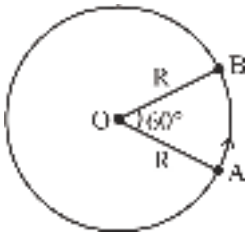
(1) 3 sec and 2000 m

(2) 5 sec and 500 m

(3) 8 sec and 1500 m

(4) 9 sec and 1800 m

34) A body moves along the circular path as shown in figure. Find magnitude of displacement between point A & B :-



(1) R/2

(2) R/4

(3) R

(4) 2R

35) The equation of motion of a projectile is: $y = 12x - \frac{5}{9}x^2$. The horizontal component of velocity is 3 ms^{-1}. Given that g = 10 ms^{-2}, what is the range of the projectile?

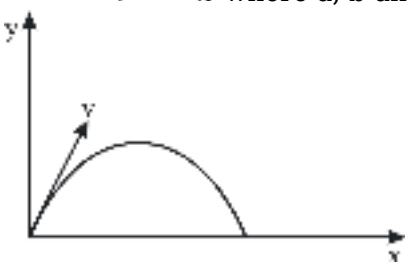
(1) 12.4 m

(2) 21.6 m

(3) 30.6 m

(4) 36.0 m

36) If the instantaneous velocity of a particle projected as shown in figure is given by $\vec{v} = a\hat{i} + (b - ct)\hat{j}$ where a, b and c are positive constants, the range on the horizontal plane will be :



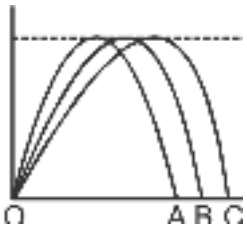
(1) 2ab/c

(2) ab/c

(3) ac/b

(4) a/2bc

37) Three projectiles A, B and C are thrown from the same point in the same plane. Their trajectories are shown in the figure. Which of the following statement is correct ?



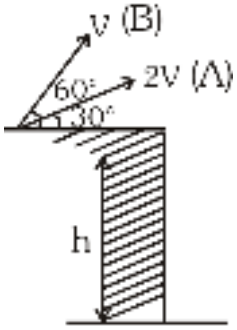
(1) The time of flight is largest for particle C

(2) The launch speed is same for all the three

(3) The vertical component of velocity is same for all the three

(4) The horizontal component of velocity is same for all the three

38) Two particles A & B are projected from a building. A is projected with speed 2V making an angle 30° with horizontal & B with speed V making an angle 60° with horizontal. Which particle will achieve greater height



(1) Particle A

(2) Particle B

(3) Particle A & B will hit at same time.

(4) None of these

39) The wrong unit conversion among the following is :

(1) 1 angstrom = 10^{-10} m

(2) 1 fermi = 10^{-15} m

(3) 1 light year = 9.46×10^{15} m

(4) 1 astronomical unit = 1.496×10^{-11} m

40) Which of the following is incorrect statement

(1) A dimensionally correct equation may be correct.

(2) A dimensionally correct equation may be incorrect.

(3) A dimensionally incorrect equation may be correct.

(4) A dimensionally incorrect equation is incorrect.

41) If force F, length L and time T be considered fundamental units, then units of mass will be :

(1) $[FLT^{-2}]$
(2) $[FL^{-2}T^{-1}]$
(3) $[FL^{-1}T^{2}]$
(4) $[F^{2}LT^{-2}]$

42)

In a particular system the units of length mass and time are chosen to be 10 cm, 10 g and 0.1 s respectively. The unit of force in this system will be equal to :

(1) 0.1 N
(2) 1 N
(3) 10 N
(4) 100 N

43) One centimetre on the main scale of vernier callipers is divided into ten equal parts. If 10 divisions of vernier scale coincide with a 8 small divisions of the main scale, the least count of the callipers is :

(1) 0.01 cm
(2) 0.02 cm
(3) 0.05 cm
(4) 0.005 cm

44) The period of oscillation of a simple pendulum in the experiment is recorded as 2.63s, 2.56s, 2.42s, 2.71s and 2.80s respectively. The average absolute error is :

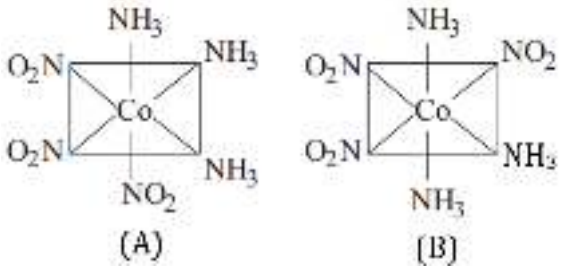
(1) 0.1s
(2) 0.11s
(3) 0.01s
(4) 1.0s

45) If L = 2.331 cm, B = 2.1 cm, then L + B = ... ? (upto correct significant figure.)

(1) 4.431 cm
(2) 4.43 cm
(3) 4.4 cm
(4) 4 cm

CHEMISTRY

1) Identify the facial (fac) and meridional (mer) isomers from the following.



(1) A → mer- ; B → mer-

(2) A → fac- ; B → fac-

(3) A → mer- ; B → fac-

(4) A → fac- ; B → mer-

2) The IUPAC name of the complex $[Co(NH_3)_4Cl_2]Cl$ is

(1) tetraamminedichlorocobalt(III)chloride

(2) tetraamminedichlorocobalt(IV)chloride

(3) tetraamminedichlorocobalt(II)chloride

(4) dichlorotetraamminecobalt(II)chloride

3) The primary and secondary valencies of chromium in the complex ion, dichlorodioxalatochromate(III) ion respectively are

(1) 3 and 4

(2) 4 and 3

(3) 6 and 3

(4) 3 and 6

4) The correct statement on the isomerism associated with the following complex ions,
I. $[Ni(H_2O)_5NH_3]^{2+}$
II. $[Ni(H_2O)_4(NH_3)_2]^{2+}$
III. $[Ni(H_2O)_3(NH_3)_3]^{2+}$

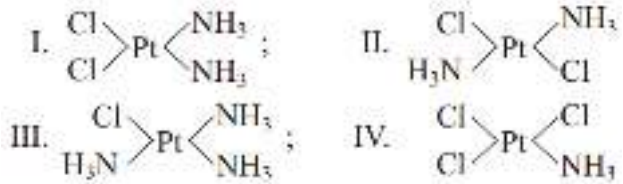
(1) I and II show geometrical and optical isomerism

(2) II and III show only optical isomerism

(3) II and III show only geometrical isomerism

(4) II and III show geometrical and optical isomerism

5) Identify the geometrical isomers of $[Pt(NH_3)_2Cl_2]$



Choose the correct option.

(1) I and II

(2) II and III

(3) I and III

(4) II and IV

6) The pair having the same magnetic moment is [atomic number, Cr=24,Mn=25,Fe=26 and Co=27]

(1) $[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$

(2) $[Mn(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{2+}$

(3) $[CoCl_4]^{2-}$ and $[Fe(H_2O)_6]^{2+}$

(4) $[Cr(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$

7) In the following questions, a statement of Assertion (A) is followed by a corresponding statement of Reason (R). Of the following statements, choose the correct one

Assertion (A):- Mohr's salt gives NH_4^+, Fe^{2+} and SO_4^{2-} ions in aqueous solution.
Reason (R):- Mohr's salt is a double salt.

(1) Both A and R are correct; R is the correct explanation of A.

(2) Both A and R are correct; R is not the correct explanation of A.

(3) A is correct; R is incorrect.

(4) R is correct; A is incorrect.

8) $[Ni(CN)_4]^{2-}$, $[MnBr_4]^{2-}$ and $[CoF_6]^{3-}$, geometry, hybridisation and magnetic moment of the ions respectively, are

(1) tetrahedral, square planar, octahedral
sp^3, dsp^2, sp^3d^2 : 5.9, 0, 4.9

(2) tetrahedral, square planar, octahedral
dsp^2, sp^3, sp^3d^2 : 0, 5.9, 4.9

(3) square planar, tetrahedral, octahedral
dsp^2, sp^3, d^2sp^3 : 5.9, 4.9, 0

(4) square planar, tetrahedral, octahedral
dsp^2, sp^3, sp^3d^2 : 0, 5.9, 4.9

9) In the following questions, a statement of Assertion (A) is followed by a corresponding statement of Reason (R). Of the following statements, choose the correct one.

Assertion (A):- Primary and secondary are the two types of valencies shown by coordination compounds.
Reason (R):- The ions/groups bound by the secondary valency to the metal have characteristic spatial arrangements corresponding to different coordination numbers.

(1) Both A and R are correct; R is the correct explanation of A.

(2) Both A and R are correct; R is not the correct explanation of A.

(3) A is correct; R is incorrect.

(4) R is correct; A is incorrect.

10)

Among the ligands NH_3, en, CN^- and CO, the correct order of their increasing field strength, is

(1) $CO < NH_3 < en < CN^-$

(2) $NH_3 < en < CN^- < CO$

(3) $CN^- < NH_3 < CO < en$

(4) $en < CN^- < NH_3 < CO$

11) Metal-carbon bond in metal carbonyls possesses

(1) only σ-character

(2) only π-character

(3) Both σ and π-character

(4) Neither σ nor π-character

12) The order of energy absorbed which is responsible for the color of complexes
(A) $[Ni(H_2O)_2(en)_2]^{2+}$
(B) $[Ni(H_2O)_4(en)]^{2+}$ and
(C) $[Ni(en)_3]^{2+}$
is

(1) (A) > (B) > (C)

(2) (C) > (B) > (A)

(3) (C) > (A) > (B)

(4) (B) > (A) > (C)

13) Increasing order of EAN of the metal in
$[Ni(CN)_4]^{2-}$, $[Fe(CN)_6^{4-}]$ and $[Cu(CN)_4]^{3-}$

(1) $[Ni(CN)_4]^{2-} < [Fe(CN)_6^{4-}] < [Cu(CN)_4]^{3-}$

(2) $[Ni(CN)_4]^{2-} < [Fe(CN)_6^{4-}] = [Cu(CN)_4]^{3-}$

(3) $[Ni(CN)_4]^{2-} = [Fe(CN)_6^{4-}] < [Cu(CN)_4]^{3-}$

(4) $[Cu(CN)_4]^{3-} < [Fe(CN)_6^{4-}] < [Ni(CN)_4]^{2-}$

14) Which of the following are paramagnetic ?
A. $[Ni(Cl)_4]^{2-}$ B. $Ni(CO)_4$ C. $[Ni(CN)_4]^{2-}$ D. $[Ni(H_2O)_6]^{2+}$ E. $Ni(PPh_3)_4$
Choose the correct answer from the options given below:

(1) A and D only

(2) A, D and E only

(3) A and C only

(4) B and E only

15) Identify the homoleptic complex(es) that is/are low spin.
(A) $[Fe(CN)_5NO]^{2-}$ (B) $[CoF_6]^{3-}$
(C) $[Fe(CN)_6]^{4-}$ (D) $[Co(NH_3)_6]^{3+}$
(E) $[Cr(H_2O)_6]^{2+}$

Choose the correct answer from the options given below :

(1) (B) and (E) only

(2) (A) and (C) only

(3) (C) and (D) only

(4) (C) only

16) Which complex is most stable ?
Where K_d = Dissociation constant

(1) $[Cu(CN)_2]^-$; $K_d = 1 \times 10^{-16}$

(2) $[Fe(CN)_6]^{-4}$; $K_d = 1 \times 10^{-37}$

(3) $[Fe(CN)_6]^{-3}$; $K_d = 1 \times 10^{-44}$

(4) $[Ag(CN)_2]^-$; $K_d = 1 \times 10^{-20}$

17) Which is inner orbital and low spin complex ?

(1) $[NiCl_4]^{-2}$

(2) $[Ni(CN)_4]^{-2}$

(3) $[CoF_6]^{-3}$

(4) $[FeF_6]^{-3}$

18)

In Ziese's salt, (C=C) bond length is :-
Note : C–C bond length in ethane is 1.54Å
C=C bond length in ethene is 1.34Å

C≡C bond length in ethyne is 1.20Å

(1) 1.37 Å

(2) 1.19 Å

(3) 1.87Å

(4) 1.34Å

19) Indicate the complex ion which shows geometrical isomerism

(1) $[Cr(H_2O)_4Cl_2]^+$

(2) $[Pt(NH_3)_3Cl]^{+1}$

(3) $[Co(NH_3)_6]^{+3}$

(4) $[Co(CN)_5\ (NC)]^{-3}$

20) Which of the following has least conductivity in aqueous solution.

(1) $[Co(NH_3)_4Cl_2]Cl$

(2) $[Co(NH_3)_3Cl_3]$

(3) $[Co(NH_3)_5Cl]Cl_2$

(4) $[Co(NH_3)_6]Cl_3$

21) Oxidation state of Fe in $Fe_4[Fe(CN)_6]_3$ is/are -

I II

(1) +1 and +2

(2) +3 and +2

(3) +3 and +4

(4) zero and zero

22) If $P > \Delta_0$ then electronic configuration of d^5 in octahedral complex

(1) $t_{2g}^5 e_g^0$

(2) $t_{2g}^3 e_g^2$

(3) $t_{2g}^4 e_g^1$

(4) $e_g^5 t_{2g}$

23) Which of the following will not act as a chelating ligand?

(1) NH_2–NH_2

(2) NH_2–CH_2–CH_2–NH_2

(3) $EDTA^{4-}$

(4) dmg

24) How can an electrochemical cell be converted into an electrolytic cell ?

(1) Applying an external opposite potential greater than E°cell

(2) Exchanging the electrodes at anode and cathode.

(3) Applying an external opposite potential lower than E°cell

(4) Reversing the flow of ions in salt bridge.

25) What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M $ZnSO_4$?

(1) The copper metal will dissolve and zinc metal will be deposited.

(2) The copper metal will dissolve with the evolution of hydrogen gas.

(3) The copper metal will dissolve with the evolution of oxygen gas.

(4) No reaction will occur.

26) The molar conductivity of a weak electrolyte when plotted against the square root of its concentration, which of the following is expected to be observed?

(1) Molar conductivity increases sharply with increase in concentration.

(2) Molar conductivity decreases sharply with increase in concentration.

(3) A small increase in molar conductivity is observed at infinite dilution.

(4) A small decrease in molar conductivity is observed at infinite dilution.

27) A solution of aluminium chloride is electrolysed for 30 minutes using a current of 2 A . The amount of aluminium deposited at the cathode is _____.

(1) 1.660 g

(2) 1.007 g

(3) 0.336 g

(4) 0.441 g

28) The standard reduction potential values of some of the p-block ions are given below. Predict the one with the strongest oxidising capacity.

(1) $E^{o}_{Sn^{4+}/Sn^{2+}} = +1.15\ V$

(2) $E^{o}_{Al^{3+}/Al} = -1.66\ V$

(3) $E^{o}_{Pb^{4+}/Pb^{2+}} = +1.67\ V$

(4) $E^{o}_{Tl^{3+}/Tl} = +1.26\ V$

29) A conducting cell is filled with 0.1 M NaCl solution. Its resistance is found to be 100 Ω whereas its conductivity is 10^{-4} Scm^{-1}. When the same cell is filled with 0.01 M KCl solution, its resistance is found to be 50 Ω. Calculate molar conductance of 0.01M KCI solution.

(1) 2 Scm^2/mol

(2) 20 Scm^2/mol

(3) 200 Scm^2/mol

(4) 100Scm^2/mol

30) From the following select the one which is **not** an example of corrosion.

(1) Rusting of iron object

(2) Production of hydrogen by electrolysis of water

(3) Tarnishing of silver

(4) Development of green coating on copper and bronze ornaments

31) Given below are two statements :
1M aqueous solutions of each of $Cu(NO_3)_2$, $AgNO_3$, $Hg_2(NO_3)_2$, $Mg(NO_3)_2$ are electrolysed using inert electrodes.
Given: $E^{0}_{Ag^{+}/Ag} = 0.80\ V$ $E^{0}_{Hg_2^{2+}/Hg} = 0.79\ V$, $E^{0}_{Cu^{2+}/Cu} = 0.24\ V$ and $E^{0}_{Mg^{2+}/Mg} = -2.37\ V$.
Statement (I) : With increasing voltage, the sequence of deposition of metals on the cathode will be Ag, Hg and Cu.
Statement (II) : Magnesium will not be deposited at the cathode instead oxygen gas will be evolved at the cathode.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both **Statement I** and **Statement II** are incorrect

(2) **Statement I** is incorrect but **Statement II** is correct

(3) **Statement I** is correct but **Statement II** is incorrect

(4) Both **Statement I** and **Statement II** are correct

32) The number of correct statements from the following is _____.
(A) Conductivity always decreases with decrease in concentration for both strong and weak electrolytes.
(B) The number of ions per unit volume that carry current in a solution increases on dilution.
(C) Molar conductivity increases with decrease in concentration.
(D) The variation in molar conductivity is different for strong and weak electrolytes
(E) For weak electrolytes, the change in molar conductivity with dilution is due to decrease in degree of dissociation.

(1) 2

(2) 3

(3) 4

(4) 5

33) In a typical fuel cell, the reactants (R) and product (P) are :-

(1) R = $H_{2(g)}$, $O_{2(g)}$; P = H_2O_2(□)

(2) R = $H_{2(g)}$, $O_{2(g)}$; P = $H_2O_{(□)}$

(3) R = $H_{2(g)}$, $O_{2(g)}$, $Cl_{2(g)}$; P = $HClO_{4(aq)}$

(4) R = $H_{2(g)}$, $N_{2(g)}$; P = $NH_{3(aq)}$

34) $\Lambda^0_{m\ (NH_4OH)}$ is equal to ____.

(1) $\Lambda^0_{m\ (NH_4OH)} + \Lambda^0_{m\ (NH_4Cl)} - \Lambda^0_{(HCl)}$

(2) $\Lambda^0_{m\ (NH_4Cl)} + \Lambda^0_{m\ (NaOH)} - \Lambda^0_{(NaCl)}$

(3) $\Lambda^0_{m\ (NH_4Cl)} + \Lambda^0_{m\ (NaCl)} - \Lambda^0_{(NaOH)}$

(4) $\Lambda^0_{m\ (NaOH)} + \Lambda^0_{m\ (NaCl)} - \Lambda^0_{(NH_4Cl)}$

35) Match the terms given in Column I with the units given in Column II.

Column I	**Column II**
(i) Λ_m	**(a)** Scm^{-1}
(ii) E_{Cell}	**(b)** m^{-1}
(iii) κ	**(c)** $S\ cm^2\ mol^{-1}$
(iv) G*	**(d)** V

(1) (i) → (c) (ii) → (d) (iii) → (b) (iv) → (a)

(2) (i) → (d) (ii) → (c) (iii) → (a) (iv) → (b)

(3) (i) → (c) (ii) → (d) (iii) → (a) (iv) → (b)

(4) (i) → (d) (ii) → (c) (iii) → (b) (iv) → (a)

36) Given below are two statements :
Statement-1: The SRP of three metallic ions A^+, B^{2+}, C^{3+} are –0.3, –0.5, 0.8 volt respectively, so oxidising power of ions is $C^{3+} > A^+ > B^{2+}$.
Statement-2: Higher the SRP, higher the oxidising power.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement-1 and Statement-2 both are True.

(2) Statement-1 is True, Statement-2 is False.

(3) Statement-1 is False, Statement-2 is True.

(4) Statement-1 and Statement-2 both are False.

37)

Consider the following relations for emf of an electrochemical cell.
(1) EMF of cell = (oxidation potential of anode) – (Reduction potential of cathode)
(2) EMF of cell = (Oxidation potential of anode) + (Reduction potential of cathode)
(3) EMF of cell = (Reductional potential of anode) + (Reduction potential of cathode)
(4) EMF of cell = (Oxidation potential of anode) – (Oxidation potential of cathode)
Which of the above relations are correct?

(1) 3 and 1

(2) 1 and 2

(3) 3 and 4

(4) 2 and 4

38) A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 10 and by passing hydrogen gas around the platinum wire at one atm pressure. The oxidation potential of electrode would be?

(1) 0.59 V

(2) 0.118 V

(3) 1.18 V

(4) 0.059 V

39) The equilibrium constant of the reaction
$Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag\ (s)$; $E° = 0.46$ V at 298 K is:

(1) 2.4×10^{10}

(2) 2.0×10^{10}

(3) 4.0×10^{10}

(4) 4.0×10^{15}

40) What does the negative value of E^0 represents ?

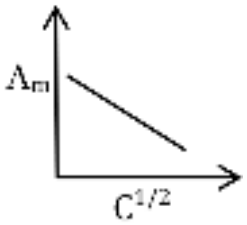
(1) Strong reducing agent then H^+/H_2 couple

(2) Strong oxidising agent then H^+/H_2 couple

(3) Weak reducing agent then H^+/H_2 couple

(4) None of these

41) For which of the following electrolyte limiting molar conductance can be obtained by the following graph-



(1) KCl

(2) CH_3COOH

(3) HCN

(4) NH_4OH

42) Pt $|H_2(1atm)|H^+(0.001M)||H^+(0.1M)|H_2(1atm)|Pt$
what will be the value of E_{cell} for this cell

(1) 0.1182 V

(2) – 0.1182 V

(3) 0.0591 V

(4) – 0.0591 V

43) A device that converts energy of combustion of fuel like hydrogen and methane directly into electrical energy is known as :-

(1) Electrolytic cell

(2) Mercury cell

(3) Ni-Cd cell

(4) Fuel cell

44) Which of the following expression correctly represents the equivalent conductance at infinite dilution of $Ca_3(PO_4)_2$ Given $\Lambda^o_{Ca^{+2}}$ and $\Lambda^o_{PO_4^{-3}}$ are the equivalent conductance at infinite dilution of the respective ions :-

(1) $\Lambda^o_{Ca^{+2}} + \Lambda^o_{PO_4^{-3}}$

(2) $\left(\Lambda^o_{Ca^{+2}} + \Lambda^o_{PO_4^{-3}}\right) \times 6$

(3) $\frac{1}{2}\Lambda^o_{Ca^{+2}} + \frac{1}{3}\Lambda^o_{PO_4^{-3}}$

(4) $3\Lambda^o_{Ca^{+2}} + 2\Lambda^o_{PO_4^{-3}}$

45) A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode ?
(Molar mass of Cu = 63.5 g/mol)

(1) 0.492g

(2) 0.325g

(3) 0.873g

(4) 0.296g

BIOLOGY

1) How many of the following are incorrect ?
(A) Flowers are morphological and embryological marvels and the sites of sexual reproduction.
(B) Much before the actual flower is seen on a plant, the decision that the plant is going to flower has taken place.
(C) Flowers have always been used as symbols for conveying important human feelings.
(D) Human beings have had a intimate relationship with flowers since time immemorial.
(E) Several hormonal and structural changes are initiated which lead to the differentiation and further development of floral primordium.
(F) Flowers are objects of aesthetic, ornamental, social, religious and cultural values.

(1) 0

(2) 1

(3) 2

(4) 3

2) The body of ovule fuses with funicle in the region called :-

(1) Raphe

(2) Integument

(3) Hilum

(4) Placenta

3) Given below are two statement. **Statement-I :-** When more than one carpel may be fused together it is called syncarpous condition. **Statement-II :-** The stigma serves as a landing platform for pollen grain.

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

4) In which of the following plants, one ovule is present in the ovary?

(1) Papaya

(2) Orchids

(3) Mango

(4) Watermelon

5) If the number of chromosomes in pollen grain are 28. Then what will be the number of chromosome in the cell of nucellus :-

(1) 14

(2) 28

(3) 56

(4) 84

6) Select the odd one :-

(1) Nucellus

(2) Funicle

(3) Chalaza

(4) Pollen grain

7) How many plant structures in the list below are haploid in nature? Polar nuclei, pollen grain, Tapetum, embryo sac, ovule, synergids, nucellus, microspore, megaspore.

(1) 6

(2) 7

(3) 5

(4) 8

8) *Amorphophallus* flower provide floral rewards to the pollinator in the form of :-

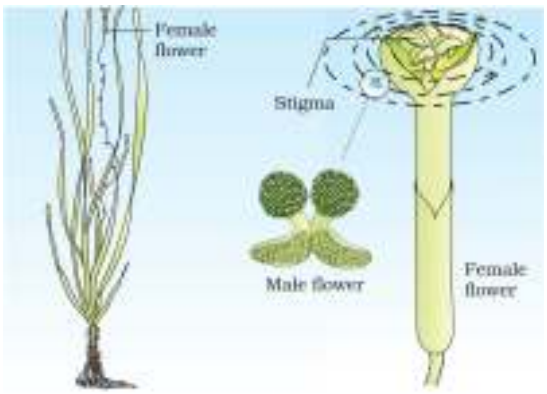
(1) Beautiful colour

(2) Providing safe place to lay eggs

(3) Nectar

(4) Providing safe place for asexual reproduction

9) The figure shows :



(1) Wind pollination in freshwater *Vallisneria*

(2) Water pollination in marine *Hydrilla*

(3) Water pollination in marine *Zostera*

(4) Water pollination in freshwater *Vallisneria*

10)

Read the following four statements (A-D) carefully:-
(A) Majority of plants use biotic agents for pollination .
(B) Wind pollination is quite common in grasses.
(C) Pollination by water is quite rare in flowering plants and is limited to about 30 genera, mostly dicotyledons.
(D) Both wind and water pollinated flowers are not very colourful and do not produce nectar.

How many of the above statements are correct ?

(1) One

(2) Two

(3) Four

(4) Three

11) Which of the following is incorrect about wind pollination ?

(1) Compact inflorescence

(2) Well – Exposed stamens

(3) Many and light pollens

(4) Ovary has many ovules

12) Which of the following statement is not correct ?

(1) Majority of insect pollinated flowers are large colourful fragrant and rich in nectar.

(2) The flowers pollinated by flies and beetles secrete foul odours to attract these animals.

(3) Pollen grains are generally sticky in animal pollinated flowers.

(4) Bees, butterflies, flies, beetles, wasps, ants, moths, birds (sunbird and humming birds) and bats are the common pollinating agents in gymnosperms.

13) Which of these is correct?

(1) Syngamy = Triple fusion + Double fertilization

(2) Double fertilization = Syngamy + Triple fusion

(3) Triple fusion = Double fertilization - Syngamy

(4) More than one option is correct

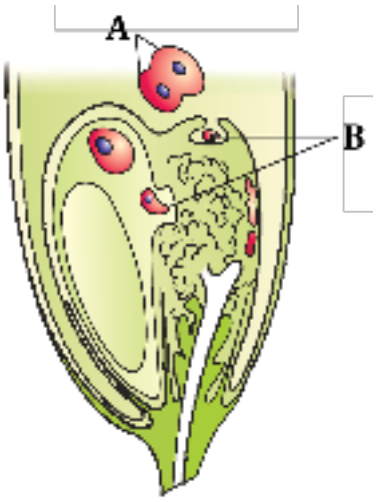
14) **Assertion** : Endosperm is a nutritive tissue and it is triploid in angiosperms.
Reason : Endosperm is formed by fusion of secondary nucleus and a male gamete. It is used by developing embryo

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false



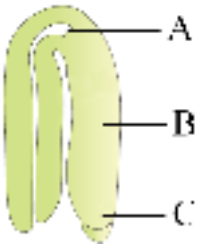
15) Identify A and B in this figure.

(1) A : Polar nuclei, B : Male gametes
(2) A : Female gametes, B : Polar nuclei
(3) A : Synergid, B : Egg cell
(4) A : Male gamets, B : Synergid

16) Which of the following group of plants is characterised by having residual, persistent nucellus ?

(1) Black pepper and Beet
(2) Wheat and Maize
(3) Sunflower and Castor
(4) *Brassica* and Barley

17) In the given diagram, labelled parts A, B and C are :



	A	B	C
(1)	Plumule	Cotyledons	Hypocotyl
(2)	Radicle	Root cap	Hypocotyl
(3)	Plumule	Hypocotyl	Radicle
(4)	Plumule	Radicle	Root cap

(1) 1
(2) 2
(3) 3
(4) 4

18) Epicotyl has a shoot apex and few leaf primordia enclosed in a hollow foliar structure known as :-

(1) Coleoptile
(2) Coleorhiza
(3) Scutellum
(4) Tigellum

19) Recently excavated seed from arctic region germinated after 10000 year is :-

(1) Lupine
(2) *Lilium*
(3) Datepalm
(4) *Phoenix dactylifera*

20) Apomictic seeds are more advantageous than hybrid seeds due to :-

(1) No seggregation of characters
(2) Maintaining hybrid vigor
(3) Increased productivity
(4) All of them

21) Fruit production without fertilisation is called ?

(1) Parthenogenesis
(2) Parthenocarpy
(3) Polyembryony
(4) Amphimixis

22) The phenomenon of occurrence of more than one embryo in a seed is known as :-

(1) Parthenocarpy
(2) Apomixis
(3) Polyembryony
(4) Parthenogenesis

23) Polyembryony is absent in :-

(1) Mango
(2) Orange
(3) Lemon
(4) China rose

24) Unisexual flowers prevent :

(1) Pollination
(2) Breeding
(3) Self-pollination
(4) Cross fertilization

25) From outer to inner, what is the sequence of wall layers in anther lobes?

(1) Epidermis, middle layers, tapetum, endothecium
(2) Epidermis, endothecium, tapetum, middle layers
(3) Epidermis, endothecium, middle layers, tapetum
(4) Tapetum, middle layers, endothecium, epidermis

26) The portion of embryonal axis between plumule (future shoot) and cotyledons is called :

(1) Hypocotyl
(2) Epicotyl
(3) Coleorhiza
(4) Coleoptile

27) In which of the following thalamus is involved in fruit formation?

(1) Apple and mango.
(2) Apple and coconut.
(3) Apple and strawberry.
(4) Apple and wheat

28) Select the correct statement from the following :
(a) Pericarp of fruit develops from integument of ovule.
(b) Apomixis is a form of sexual reproduction
(c) Orchid fruits contain one to two tiny seeds.
(d) Scutellum is the cotyledon found in embryo of grasses.

(1) a, b only
(2) b, c, d only
(3) d only
(4) a, b, c only

29) What will be ploidy of the cells of nucellus, megaspore mother cell, functional megaspore and embryosac respectively

(1) n, 2n, 3n, 4n
(2) n, 3n, 4n, 2n
(3) 2n, 2n, n, n
(4) 3n, 4n, 2n, n

30) Which of the following is not incorrect regarding the angiosperm?

(1) Embryo development preceeds the endosperm development.
(2) Endosperm is formed due to triple fusion.
(3) Endosperm provide nutrition to seedling during seed germination in exalbuminous seed.
(4) All of the above

31) The _____ lead to vas deferens that ascends to the _____ and loops over the _____.

(1) Prostate, Stomach, Urinary bladder
(2) Epididymis, Abdomen, Urinary bladder
(3) Vasa efferentia, Abdomen, Ureter
(4) Urinary bladder, Ejaculatory duct, Abdomen

32) The enlarged end of penis is covered by a loose fold of skin called :

(1) Glans penis
(2) Foreskin
(3) Hymen
(4) Urethral meatus

33) Vasa efferentia are the ductules leading from -

(1) Epididymis to urethra.
(2) Vas deferens to epididymis.
(3) Rete testis to epididymis.
(4) Testicular lobules to rete testis.

34) Ejaculation of human male contains about 200 – 300 million sperms, of which for normal fertility ____ % sperms must have normal shape and size and at least ____% must show energetic motility.

(1) 40, 60
(2) 50, 50
(3) 60, 40
(4) 30, 70

35) Which of the following stage of oogenesis forms a membrane called zona pellucida surrounding it?

(1) Oogonia
(2) Polar body
(3) Corpus luteum
(4) Secondary oocyte

36) Colostrum -

(1) Is a hormone essential for milk secretion.
(2) Can be synthesized by the newborn infant but not by a foetus.
(3) Stimulates further secretion of oxytocin for uterine contraction.
(4) Is a source of antibodies essential to develop resistance against diseases in new born babies.

37) Select the correct statements regarding oogenesis.
(i) It is initiated during the embryonic development stage when millions of oogonia are formed within each fetal ovary.

(ii) Graafian follicle releases primary oocyte from the ovary by ovulation.
(iii) At puberty only 60,000 – 80,000 primary follicles are left in each ovary.
(iv) Secondary follicle transforms in to primary follicle which is characterised by fluid filled cavity called antrum.

(1) (i), (ii) and (iii)

(2) (i) and (iii)

(3) (ii) and (iv)

(4) All the four statements

38)

Which of the following statement is correct regarding menstruation?

(1) The menstrual fluid can easily clot.

(2) The end of the cycle of menstruation is called menarche.

(3) At menopause in the female, there is especially abrupt decrease in gonadotropic hormones.

(4) In human female, menstruation can be deferred by the administration of combination of estrogen and progesterone.

39) Read the following statements (i to v) and answer the following question.
(i) This structure is also called womb.
(ii) Its shape is like an inverted pear.
(iii) The process of fertilization takes place in this structure.
(iv) The wall of this structure has three layers of tissue.
(v) It secretes several steroid hormones.
Identify the correct characteristics feature regarding uterus from the above statements.

(1) (i) and (iv)

(2) (iii) and (v)

(3) (i), (ii) and (iv)

(4) All the five statements

40) Which of the following statement regarding female reproductive system is (are) correct?
(i) Myometrium undergoes strong contraction at the time of delivery of baby.
(ii) Ovary is secondary female sex organ which produces female gamete and steroid hormones.
(iii) Ovarian stroma is divided into two zones: inner cortex and outer medulla.
(iv) Infundibulum possess finger like projections which help in collection of ovum.
(v) A functional mammary gland is the characteristic of all the mammals (including male and female).

(1) (i) and (iv)

(2) (i), (ii), (iii) and (v)

(3) (iii), (iv) and (v)

(4) All the five statements

41) **Assertion :** During fertilization, head of spermatozoa enters egg.

Reason : If several spermatozoa hit the egg at same time, all can enter the egg.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

42) **Assertion :** Corpus luteum degenerates in the absence of fertilization.
Reason : Progesterone level decreases after the degeneration of corpus luteum.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

43) In the given columns, column I contain structures of male reproductive system and column II contains its feature. Select the correct match from the options given below.

Column I (Structure of male Reproductive System		**Column II (Features)**	
A.	Seminiferous tubule	I.	Network of tubules
B.	Rete testis	II.	Secondary sexual characters
C.	Leydig cells	III.	Sperm formation occurs
D.	Prepuce	IV.	Place of implantation
		V.	Terminal skin of penis

(1) A – I; B – II; C – III; D – V

(2) A – III; B – I; C – II; D – V

(3) A – III; B – I; C – IV; D – II

(4) A – II; B – IV; C – III; D – V

44) Which of the following pair is incorrectly matched?

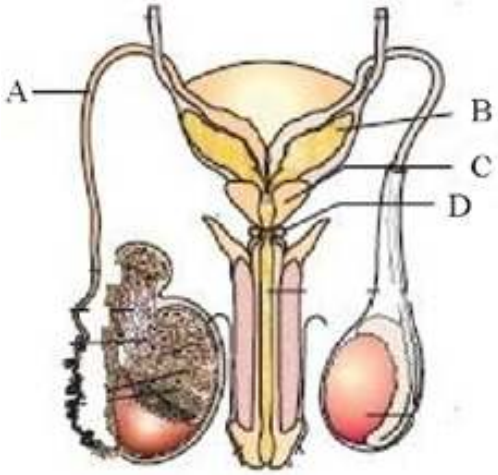
(1) Leydig cells – Testosterone

(2) Spermatogenesis – Seminiferous tubules

(3) Male reproductive system – Pelvis region

(4) Primary spermatocyte – Mitotic division

45) Given below is a diagrammatic sketch of a portion of human male reproductive system. Select the correct set of the names of the parts marked as A, B, C, and D respectively.



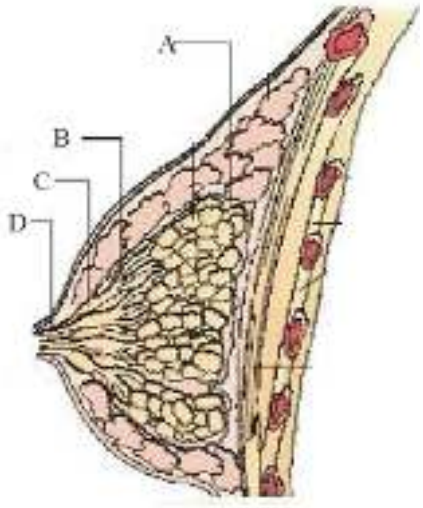
(1) A-Ureter, B-Seminal vesicle, C-Prostate, D-Bulbourethral gland

(2) A-Ureter, B-Prostate, C-Seminal vesicle, D-Bulbourethral gland

(3) A-Vas deferens, B-Seminal vesicle, C-Prostate, D-Bulbourethral gland

(4) A-Vas deferens, B-Seminal vesicle, C-Bulbourethral gland, D-Prostate deferens

46) Refer the figure of mammary gland with few structures marked as A, B, C and D. Which structure contains clusters of milk secreting cells?



(1) A

(2) B

(3) C

(4) D

47) **Assertion :** Testicular lobules are the compartments present in testis.
Reason : These lobules are involved in the process of fertilization.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

48) In the given columns, column-I contain structures of female reproductive system and column-II contain its feature.
Select the correct match from the option given below.

Column I (Structures of female reproductive system)		Column II (Features)	
A.	Ampulla	I.	It undergoes cyclical changes during menstrual cycle.
B.	Labia majora	II.	It helps in collection of ovum after ovulation.
C.	Oviduct	III.	Wider part of fallopian tube where fusion of male and female gametes takes place.
D.	Fimbriae	IV.	Larger hairy folds which extend down from the mons pubis and surrounds the vaginal opening.
E.	Endometrium	V.	Also called fallopian tubes, which extend from the periphery of each ovary to the womb.

(1) A – I; B – II; C – III; D – V; E – IV

(2) A – III; B – I; C – II; D – V; E – IV

(3) A – III; B – IV; C – V; D – II; E – I

(4) A – II; B – IV; C – III; D – V; E – I

49) In the given columns, column-I contain various phases of menstrual cycle and column -II contain its features.
Select the correct match from the options given below.

	Column I (Phases of menstrual cycle)		Column II (Features)
A.	Menstrual phase	I.	Breakdown of endometrial lining of uterus along with its blood vessels which form liquid that comes out of vagina
B.	Luteal phase	II.	A temporary endocrine structure is formed and secretes a hormone which maintains endometrium and implantation of fertilized ovum and events of pregnancy.
C.	Follicular phase	III.	Secretion of luteinizing hormone at its maximum level and induces breakdown of mature follicle to release the female gamete.
D.	Ovulatory phase	IV.	Formation of mature Graafian follicle and regeneration of endometrium of uterus.

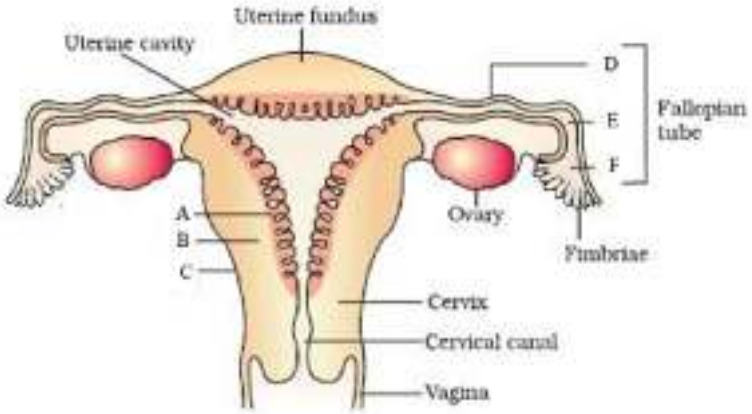
(1) A – IV; B – II; C – III; D – I

(2) A – III; B – I; C – II; D – IV

(3) A – III; B – I; C – IV; D – II

(4) A – I; B – II; C – IV; D – III

50) The given figure shows the diagrammatic sectional view of female reproductive system with few structures marked as A, B, C, D, E and F.



Which of the following options shows the correct labeling of A–F?

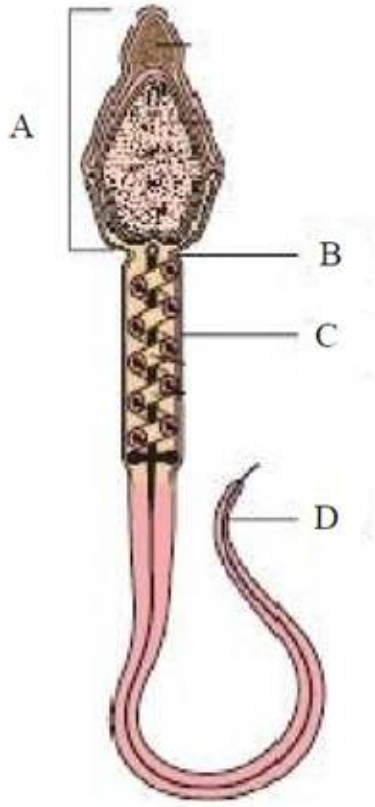
(1) A → Myometrium, B → Isthmus, C → Endometrium, D → Perimetrium, E → Ampulla, F → Infundibulum

(2) A → Infundibulum, B → Perimetrium, C → Endometrium, D → Myometrium, E → Ampulla, F → Isthmus

(3) A → Endometrium, B → Myometrium, C → Perimetrium, D → Isthmus, E → Ampulla, F → Infundibulum

(4) A ® Perimetrium, B ® Endometrium, C ® Isthmus, D ® Infundibulum, E ® Ampulla, F ® Myometrium

51) The figure given below shows the structure of sperm. Identify the correct feature corresponding to the marked structure A, B, C and D.



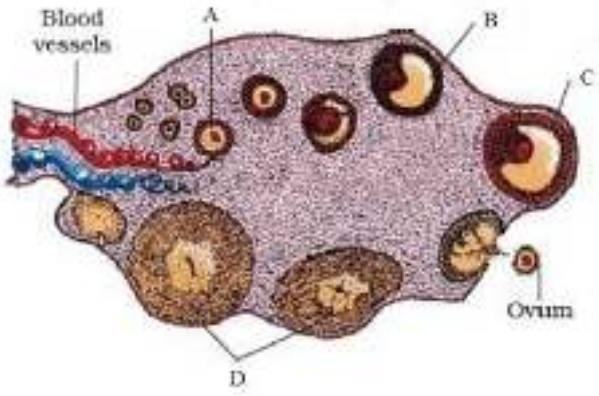
(1) A – Head: Its anterior portion is covered by a structure filled with enzymes that help in the fusion of male and female gametes.

(2) B – Middle piece: It contains a haploid nucleus.

(3) C – Neck: It possesses few ribosomes which produces energy for the process of fertilization.

(4) D – Tail: It releases energy source for swimming of sperm.

52) The figure given below shows the sectional view of ovary. Select the option which gives correct identification of marked structure (A to D) and its feature.



(1) A : Primary follicle, it is also called gamete mother cell.

(2) B : Corpus luteum, it cannot be formed and added after birth.

(3) C : Graafian follicle, mature follicle which ruptures to release secondary oocyte.

(4) D : Tertiary follicle, a large number of this follicle degenerates during the phase from birth to puberty.

53) Spermatids are transformed into sperm by a process called_____.

(1) Spermiation

(2) Implantation

(3) Insemination

(4) Spermiogenesis

54) Increased secretion of which hormone start the process of sperm formation at the time of puberty?

(1) GH

(2) TSH

(3) PRL

(4) GnRH

55) Which of the following is a finger like structure and lies at the upper junction of the two labia minora above the urethral opening?

(1) Clitoris

(2) Oviduct

(3) Ampulla

(4) Chorionic villi

56) Semen is a constituent of seminal plasma with _____.

(1) Ovum
(2) Sperms
(3) Zygote
(4) Follicle

57) Which of the following contains a fluid filled cavity called antrum?

(1) Primary spermatocyte
(2) Primary follicle of ovary
(3) Tertiary follicle of ovary
(4) Secondary spermatocyte

58) Menstruation is triggered by a sudden decline in the amount of hormone secreted by corpus luteum.
Identify the hormone -

(1) Luteinizing hormone
(2) Follicle stimulating hormone
(3) Progesterone
(4) Estrogen

59) The embryo with 8 to 16 blastomeres is called _____.

(1) Zygote
(2) Morula
(3) Placenta
(4) Blastocysts

60) Placenta acts as an -

(1) Embryo
(2) Corpus luteum
(3) Exocrine gland
(4) Endocrine tissue

61) Which of the following statement is incorrect?

(1) According to 2011 census our population growth rate was less than 2%.
(2) Marriageable age for male and female is respectively 21 and 18 years.
(3) An ideal contraceptive should be reversible.
(4) The problem of infertility in India lies most often in female partner.

62) Which of the following statement regarding natural methods of contraception is true?

(1) They increase phagocytosis of sperms.
(2) They employ barriers to prevent fertilization.

(3) They are the natural ways of avoiding chances of fertilization.

(4) They are surgical and terminal methods.

63) Read the following statements and mark the correct option -
(i) MTP was legalized in 1971.
(ii) Inability to conceive or produce children even after 2 years of unprotected sexual cohabitation is called infertility.
(iii) Surgical method of contraception prevents gamete formation.
(iv) MTPs are relatively safe up to 14 weeks of pregnancy.

(1) (i) and (ii)

(2) (ii) and (iii)

(3) (iii) and (iv)

(4) (i) and (iii)

64) According to which of the following organization "reproductive health means a total well-being in all aspects of reproduction"?

(1) WHL

(2) UNESCO

(3) WHO

(4) WWW

65) Which one amongst the following is the first country in the world to initiate action plans and programmes at a national level to attain total reproductive health as social goal?

(1) China

(2) India

(3) Japan

(4) USA

66) The family planning programmes in India were initiated in -

(1) 1951

(2) 1961

(3) 1971

(4) 1981

67) RCH stands for -

(1) Routine Check-up of Health

(2) Reproduction Cum Hygiene

(3) Reversible Contraceptive Hazards

(4) Reproductive and Child Health Care

68) ‘Saheli’ a new oral contraceptive developed by -

(1) All Indian Institute of Medical Science
(2) Central Drug Research Institute
(3) Health Care Pvt. Ltd.
(4) Bharat Immunological & Biologicals corp. Ltd.

69) What is the marriageable age for the females and males respectively in India?

(1) 18, 18
(2) 18, 25
(3) 21, 18
(4) 18, 21

70) Which of the following contraceptive also provides protection from contacting STDs and AIDS ?

(1) Diaphragms
(2) Spermicidal foams
(3) Condoms
(4) Lactational amenorrhoea

71) Which of the following groups of contraceptives are included under barrier methods?

(1) Condom, Vaults, IUDs, Cervical caps
(2) IUDs, Diaphragm, Abstinence, Injections
(3) Condom, Diaphragm, Cervical caps, Vaults
(4) Surgical methods, Oral contraceptives, Cervical caps, vaults

72) IUDs stands for -

(1) Intra Uterine Devices
(2) Internal Uterine Devices
(3) Inseminated Uterine Devices
(4) Injected Uterine Devices

73) Artificial insemination mean -

(1) Transfer of sperms of husband to a test tube containing ova.
(2) Artificial introduction of sperms of a healthy donor into the vagina.
(3) Introduction of sperms of a healthy donor directly into the ovary.
(4) Transfer of sperms of a healthy donor to a test tube containing ova.

74) Progestasert and LNG-20 are -

(1) Implants
(2) Copper releasing IUDs
(3) Non-medicated IUDs
(4) Hormone releasing IUDs

75) The most important component of the oral contraceptive pills is -

(1) Progesterone
(2) Growth hormone
(3) Thyroxine
(4) Luteinizing hormone

76) Tying up or removing a small part of fallopian duct is called -

(1) Vasectomy
(2) Ductus arteriosus
(3) Orchidectomy
(4) Tubectomy

77) In which year, MTP was legalised in India?

(1) 1971
(2) 1951
(3) 1981
(4) 1923

78) MTPs are considered relatively safe during the first _____ weeks of pregnancy.

(1) 12
(2) 15
(3) 18
(4) 20

79) To form embryo in vitro, the male gamete is transferred into female gamete directly. Such technique is called_____.

(1) IUI
(2) IUT
(3) ICSI
(4) GIFT

80) The transfer of zygote or early embryo (up to 8 blastomeres) into fallopian tube is called _____.

(1) IVF and ET
(2) ZIFT
(3) GIFT
(4) IUT

81) The Test-tube Baby Programme employs which one of the following embryo transfer techniques?

(1) Intra Cytoplasmic Sperm Injection (ICSI)

(2) Intra Uterine Insemination (IUI)

(3) Gamete Intra Fallopian Transfer (GIFT)

(4) Zygote Intra Fallopian Transfer (ZIFT)

82) **Assertion :** Only mother should be blamed for the birth of the girls in the family.
Reason : Father is not responsible for the sex of the child.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

83) **Assertion :** Cu-T and Cu-7 suppress sperm motility.
Reason : Hormones released by them affect sperm motility.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

84) **Assertion :** Amniocentesis is often misused.
Reason : Amniocentesis is meant for determining the genetic disorders in the foetus, but it is being used to determine the sex of the foetus, to kill the normal female foetus.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

85) Given below are four methods (A-D) of contraceptive in column-I and their modes of action in achieving contraception in column-II. Match the columns and select the correct option :

	Column – I (Method)		**Column – II (Mode of Action)**
A.	The contraceptive pill	I.	Prevents sperms reaching cervix
B.	Condom	II.	Prevents implantation
C.	Vasectomy	III.	Prevents ovulation
D.	IUDs	IV.	Semen contains no sperms

(1) A – II; B – III; C – I; D – IV

(2) A – III; B – I; C – IV; D – II

(3) A – IV; B – I; C – II; D – III

(4) A – III; B – IV; C – I; D – II

86) Which of the following option is correct regarding the diagram given below?



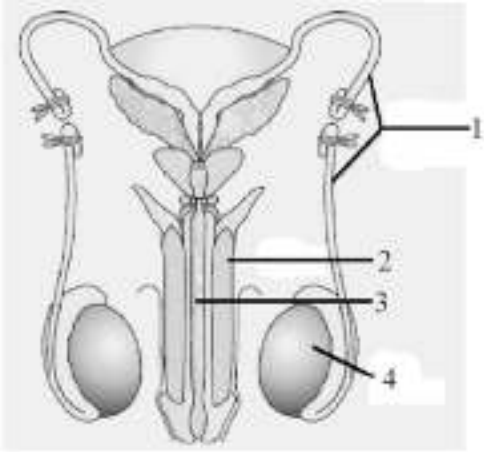
(1) It is a device made of rubber and inserted into the female reproductive tract to cover the cervix during coitus.

(2) It is a device made of thin rubber/ latex sheath and are used to cover penis in the male.

(3) This device is inserted by doctors in the uterus through vagina and increases phagocytosis of sperms within the uterus.

(4) It is a set of 6-small plastic capsules (called implant) which are placed under the skin of a women's upper arm and it prevent pregnancy.

87) In the given figure which marked number (1, 2, 3, 4) is tied and cut to block fertilization



(1) 1

(2) 2

(3) 3

(4) 4

88) The given figure shows one of the elements releasing intrauterine device. Select the option which shows the correct identification of the device and its feature.



(1) CuT; suppress sperm motility and its fertilizing capacity.

(2) Cu7; make uterus unsuitable for the attachment of blastocysts.

(3) Lippes loop; protect the users from contracting AIDS and STDs.
(4) LNG - 20; acts as spermicidal means and decrease the contraceptive efficiency.

89) On which days of the menstrual cycle is ovulation expected?

(1) 10th - 30th
(2) 1st - 10th
(3) 10th - 17th
(4) 18th - 25th

90) The success of birth control programmes in controlling population growth is dependent on -

(1) Use of contraceptives
(2) Tubectomy
(3) Vasectomy
(4) Acceptability of the above by the people

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	1	2	2	3	4	3	1	1	4	3	1	1	1	4	3	1	3	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	2	3	4	1	1	2	2	1	4	2	4	3	2	1	3	1	4	3
Q.	41	42	43	44	45															
A.	3	1	2	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	1	4	3	1	1	1	4	2	2	3	3	2	1	3	3	2	1	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	2	1	1	4	2	3	3	2	2	3	2	2	2	3	1	4	1	4	1
Q.	86	87	88	89	90															
A.	1	1	4	1	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	3	3	3	4	1	2	4	4	4	4	4	1	1	1	3	1	1	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	4	3	3	2	3	3	3	2	2	2	3	3	4	4	2	4	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	2	4	3	1	3	3	4	3	1	3	4	4	1	2	3	3	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	1	3	2	1	4	2	4	3	3	1	2	4	1	4	1	1	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	3	1	2	4	1	1	3	4										

SOLUTIONS

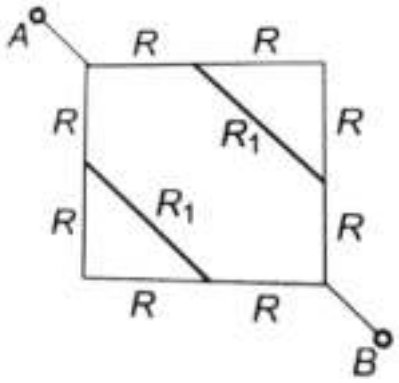
PHYSICS

1) **Question Explaination:** Unknown resistance in parallel.
Concept: Ohm's Law & Kirchhoff's Laws.
Solution: In a parallel circuit, the voltage (V) across all branches is the same.
1. Find Voltage: Using the 20Ω resistor, $V = I \times R = 0.3\ A \times 20\Omega = 6\ V$.
2. Find Current in 15Ω branch: $I_{15} = \frac{V}{R} = \frac{6\ V}{15\Omega} = 0.4\ A$.
3. Find Current in R_1 branch: Total current is 0.8 A.
$I_{R1} = I_{total} - (I_{20} + I_{15}) = 0.8 - (0.3 + 0.4) = 0.1\ A$
4. Calculate $R_1 : R_1 = \frac{V}{I_{In}} = \frac{6\ V}{0.1\ A} = 60\Omega$.
Final Answer: Option (4)

2) **Question Explanation:** The problem asks to calculate the net electrical resistance between points P and Q for a complex wire frame consisting of an outer square and an inscribed diamond.

Concept: Ohm's Law, Resistance formula ($R = \frac{\rho L}{A}$), Series and Parallel combinations,

Solution:
The figure can be redrawn as



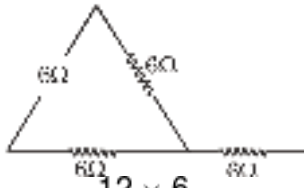
The outer square has sides of length l. Each side is split into two segments of l/2. Let the resistance of a segment of length l/2 be R :
$R = \frac{\rho(l/2)}{A} = \frac{\rho l}{2A}$
Let the resistance of one diamond side be R_1 :
$R_1 = \frac{\rho(l/\sqrt{2})}{A} = \frac{\rho l}{\sqrt{2}A} = \sqrt{2}R$
$R_{eq} = \frac{1}{2}\left[\frac{2RR_1}{2R + R_1} + 2R\right] = \frac{1}{2}\left[\frac{2R.\sqrt{2}R}{2R + \sqrt{2}R} + 2R\right]$
$R_{eq} = \sqrt{2}R$
$R_{eq} = \sqrt{2}\frac{\rho l}{2A} = \frac{\rho l}{\sqrt{2}A}$

Final Answer : Option (1)

3) **Question Explanation:**
Calculating heat produced in one branch of a parallel circuit based on known heat production in another branch.
Concept:
Heat produced per second (Power) is given by $P = I^2R$ or $P = \frac{V^2}{R}$. In a parallel circuit, the voltage (V) across the branches is the same.
Solution:
Analyze the circuit:
Bottom branch has a resistor $R_b = 5\ \Omega$.
Top branch has two resistors in series, so total resistance $R_t = 4 + 6 = 10\ \Omega$.
Find voltage (V) across branches:
Heat in $5\ \Omega$ resistor $= 10$ cal/s.
Using $P = \frac{V^2}{R} \Rightarrow 10 = \frac{V^2}{5} \Rightarrow V^2 = 50$.
Find current in top branch (I_t):
Since $V^2 = 50$ and total resistance of top branch is $10\ \Omega$:
$$I_t = \frac{V}{R_t} \Rightarrow I_t^2 = \frac{V^2}{R_t^2} = \frac{50}{10^2} = \frac{50}{100} = 0.5\ A^2$$
.
Calculate heat in $4\ \Omega$ resistor(H_4):
$H_4 = I_t^2 \times R_4$
$H_4 = 0.5 \times 4 = 2$ cal/s.
Final Answer:
Option (2)

4)

Explaining: Given: Wire is bent in a shap as shown.
Ask: The effective resistance
Concept: Sereis combination: $R_{equ} = R_1 + R_2$
Parallel combination: $\frac{1}{R_{equ}} = \frac{1}{R_1} + \frac{1}{R_2}$
Solution:



$$R_{Eq} = \frac{12 \times 6}{12 + 6} + 6 = 10\Omega$$
Final Answer: (2)

5) **Explaining:** Given: An electric circuit
Ask: Terminal potential difference
Concept: $V_t = E - ir$ (Discharging)
$V_t = E + ir$ (Charging)

Solution:

$$i = \frac{6-2}{1+3} = 1\,A$$

$V_2 = \varepsilon + ir = 2 + 1 \times 3 = 5V$

Final Answer: (3)

6) **Explaining:** Given: A conducting medium having two types of charges

Ask: Current (Net) when potential difference is applied

Concept: $I = neAV_d$

Solution:

$i = i_1 + i_2$

$$= neAV + n(2e)A\left(\frac{V}{4}\right)$$

$$= \frac{3}{2} neAV$$

Final Answer: (4)

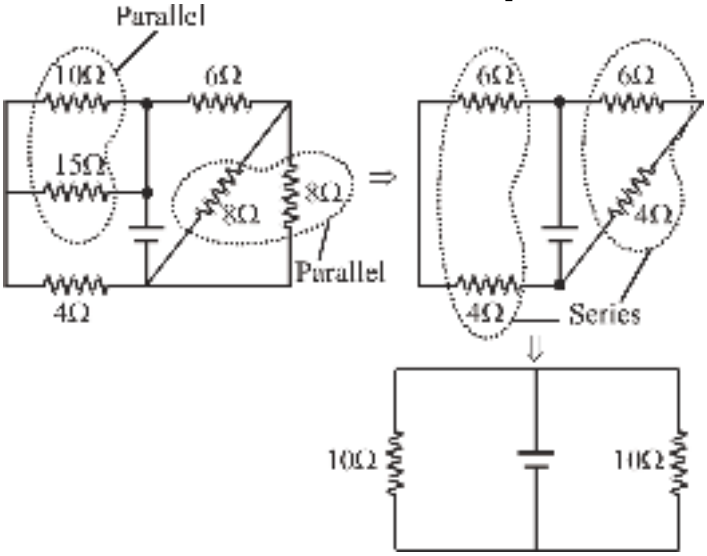
7) **Explaining:** Given: An electric circuit

Ask: Equivalent resistance across the source terminal

Concept: Series and parallel combination of resistances

Solution:

Given circuit can be reduced to a simple circuit as shown in figures below



i.e. $R_{eq} = 5\Omega$.

Final Answer: (3)

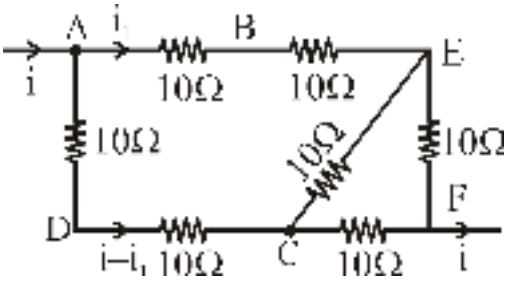
8) **Explaining:** Give: A resistive network

Ask: Current in branch AD

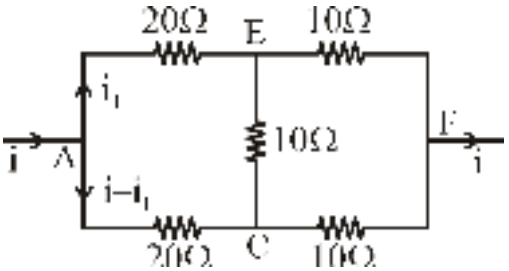
Concept: Balanced wheat stone bridge and current distribution

Solution:

This is balanced wheat stone bridge.



OR



Resistance of A to E & A to C are equal

So, $i_1 = i - i_1$

$\Rightarrow i_1 = \frac{i}{2}, \; i - i_1 = \frac{i}{2}$

This is current through AD.

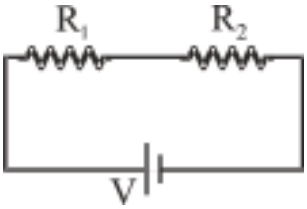
Final Answer: (1)

9) **Explaining:** Given: Two bulb connected in series

Ask: Maximum voltage that can be applied

Concept: $R_b = \frac{V_r^2}{P_r}$ and $i \le i_r$

Solution:



$V = i(R_1 + R_2)$

$i \le \frac{40}{200}, \frac{100}{200}$

$\therefore i_{max} = \frac{40}{200}$ A

$\therefore V_{max} = i_{max}(R_1 + R_2)$

Final Answer: (1)

10) **Explaining:** Given: A resistive network

Ask: Equivalent resistance

Concept: Parallel combination of resistances

Solution:

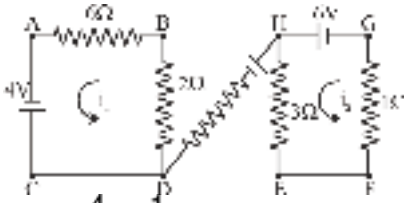
Three resistances are in parallel.

$\therefore \frac{1}{R'} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R}$

The equivalent resistance $R' = \frac{R}{3}\Omega$

Final Answer: (4)

11) **Explaining:** Given: An electric circuit
Ask: To choose the incorrect statement
Concept: KVL and KCL
Solution:



$$i_1 = \frac{4}{8} = \frac{1}{2}A$$

$$i_2 = \frac{6}{4} = \frac{3}{2}A$$

$V_H - V_D = 4V$
$V_D - V_B = 1V$
$V_H - V_B = 5V$
$V_H - V_D = 4V$ ....(i)
$V_H - V_G = 6V$ ....(ii)
Equation (ii) – (i)
$V_D - V_G = 2V$ **Final Answer: (3)**

12) **Question Explanation:** The question asks to calculate the current I flowing through the 10Ω resistor into a junction that splits into two other branches with known voltages and resistances.
Concept: Kirchhoff's Current Law (KCL), Ohm's Law, Nodal Analysis
Solution:
Let the potential at the central junction be V_J. According to Kirchhoff's Current Law (KCL), the sum of currents entering the junction must equal the sum of currents leaving it.

$$\frac{30 - V_J}{10} = \frac{V_J - 12}{20} + \frac{V_J - 2}{30}$$

$$180 - 6V_J = 3V_J - 36 + 2V_J - 4$$

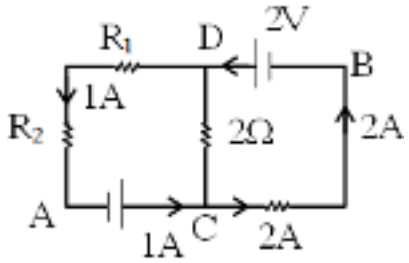
$$180 - 6V_J = 5V_J - 40$$

$$220 = 11V_J \Longrightarrow V_J = 20\ V$$

$$I = \frac{30 - 20}{10} = \frac{10}{10} = 1\ A$$

Final Answer: Option (1)

13) **Explaining:** Given: An electric circuit
Ask: Potential at point B
Concept: KVL
Solution:



Applying KCL at C indicates that current in D C will be 1A from D to C

$\therefore V_A - V_B = -1 - 1 \times 2 + 2$

(through h = –1 volt Path (ACDB)

$0 - V_B = -1$ volt

$\therefore V_B = 1$volt

Final Answer: (1)

14) **Explaining:** Given: 48 cells with E = 2V, r = 4Ω

Ask: Maximum current

Concept: Cells combination and maximum power transfer theorem

Solution:

$\frac{nr}{m} = R$ (for max. current)

$\Rightarrow \frac{n}{m}(4) = 12$

$\Rightarrow \frac{n}{m} = 3$

$\Rightarrow 3m^2 = 48$

$m = 4$

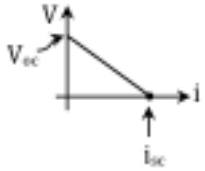
$I = \frac{nE}{\frac{nr}{M} + R} n = 12$

$= \frac{12 \times 2}{12 + 12} = \frac{24}{24} = 1\,A$

Final Answer: (1)

15) **Explaining:** Given: Graph of V vs i

Ask: Emf of each cell



Concept: Combination of cells and

Solution:

$i = \frac{10E}{10r + R} \Rightarrow V = 10E - ir$

$\frac{10E}{14} = r \Rightarrow r = \frac{5E}{7}$

$V = 10E - ir$

$\Rightarrow 6 = 10E - \frac{5E}{7} \times 12$

$6 = \frac{V_0 E}{7}$

$\Rightarrow E = \frac{21}{5} = 4.2\,V$

Final Answer: (4)

16)

Explain the Question : A screw gauge
s = 0.5 mm
$N_{rotation}$ = 1
$N_{circular}$ = 50
MSR = 2.5 mm
CSR = 45 CSD
ZE = –0.03 mm
NR = ?
Concept : This question is based on screw gauge.
Solution :

$$\text{Pitch} \Rightarrow = \frac{S}{N_{rotation}} = \frac{0.5mm}{1} = 0.5mm$$

$$\text{Least count} \Rightarrow LC = \frac{P}{N_{circular}} = \frac{0.5mm}{50} = 0.01mm$$

Net reading ⇒
NR = MSR + (CSR × LC) – ZE
= 2.5 mm + (45×0.01mm)– (–0.03mm)
=2.5 mm + 0.45 mm + 0.03 mm
= 2.98 mm
Final answer : 3

17) **Explaining :** Match the column
Concept : This question is based on ground–ground projectile motion.

$$T = \frac{2U_y}{g}$$

$$H = \frac{U_y^2}{2g}$$

$$R = \frac{2U_xU_y}{g}$$

Solution :

$$\vec{U} = U_x\hat{i} + U_y\hat{j}$$

$$\vec{U} = 30\hat{i} + 40\hat{j}$$

$$T = \frac{2U_y}{g} = \frac{2 \times 40}{10} = 8 \text{ sec.}$$

$$H = \frac{U_y^2}{2g} = \frac{40 \times 40}{20} = 80 \text{ m}$$

$$R = \frac{2U_xU_y}{g} = \frac{2(30)(40)}{10} = 240 \text{ m}$$

& Horizontal velocity remains constant.
Final Answer : (1)

18) **Explaining :** Find percentage error in density.
Concept : This question is based on propagation in error.

Solution : Density (ρ) = $\frac{\text{mass(m)}}{\text{volume(V)}}$

$\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$

$\frac{\Delta\rho}{\rho} = \left(\frac{0.05}{5}\right)100\% + \left(\frac{0.05}{1}\right)100\%$

= 1% + 5% = 6%

Final Answer : (3)

19) **Explaining :** Find velocity for given time.

Concept : This question is based on average velocity.

Solution :

Part 1: Instantaneous Velocity at t = 3.0 s Velocity (v) is the rate of change of position (x).

$x = A + Bt^2$

$v = \frac{dx}{dt} = \frac{d}{dt}(10 + 3t^2)$

$v = 0 + (2 \times 3)t =$ **6t**

Now, substitute **t = 3.0 s**:

$v = 6 \times 3.0$

v = 18 m/s

Part 2: Average Velocity between 1.0 s and 3.0 s Average Velocity = (Total Displacement) / (Total Time) $\mathbf{V_{avg} = [x(t_2) - x(t_1)] / [t_2 - t_1]}$

1. Calculate position at **t = 3.0 s** (x_2): = $10 + 3 \times (3)^2 = 10 + 3 \times 9 = 10 + 27 =$ **37 m**

2. Calculate position at **t = 1.0 s** (x_1): = $10 + 3 \times (1)^2 = 10 + 3 \times 1 =$ **13 m**

3. Calculate Average Velocity: $V_{avg} = (37 - 13) / (3.0 - 1.0)$ $V_{avg} = 24 / 2.0$

4. $\mathbf{V_{avg} = 12\ m/s}$

Final Answer : (1)

20) **Explaining :** To find v-t graph for given a - t graph

Concept : This question is based on Key point :

(i) If a is constant then v - t graph is line as

(ii) change in velocity = area under a-t graphs.

Solution : Area of a – t graph gives change in velocity

u = 0

area = 10 (for t = 0 to 2 sec)

So, velocity increases from 0 to 10 in 2 sec, linearly.

Final Answer : (2)

21) **Explaining :** Calculate displacement in given time.

Concept : This question is based on apply equation of motion.

Solution : From $S = ut + \frac{1}{2}at^2$

$S_1 = \frac{1}{2}a(p-1)^2$ and $S_2 = \frac{1}{2}ap^2$

[As u = 0]

From $S_n = u + \frac{a}{2}(2n-1)$

$S_{(p^2-p+1)^{th}} = \frac{a}{2}[2(p^2-p+1)-1]$

$S_{(p^2-p+1)^{th}} = \frac{a}{2}[2p^2 - 2p + 1]$

It is clear that $S_{(p^2-p+1)^{th}} = S_1 + S_2$

Final Answer : (1)

22) **Explaining :** Find relation b/w T, P, d & E.

Concept : Relation b/w diff. physical quantities using dimensional analysis.

Solution :

$T = KP^a d^b E^c$

$[T] = [ML^{-1}T^{-2}]^a [ML^{-3}]^b [ML^2T^{-2}]^c$

$a + b + c = 0$

$-a - 3b + 2c = 0$

$-2a - 2c = 1$

$a + c = -\frac{1}{2}$

$a = -\frac{5}{6}, b = \frac{1}{2}, c = \frac{1}{3}$

So, $T \propto P^{-5/6}\, d^{1/2}\, E^{1/3}$

Final Answer : (1)

23) **Explaining :** Find the distance of the **second drop** from the **top (the tap)** at that instant.

Concept : This question is based on equation of motion.

Solution :

Step 1: Understand the Timing Total drops involved: 5 Let the time interval between any two drops be **'t'**. When the 5th drop is at the tap (Time = 0), the intervals are: 5th drop: At t = 0

4th drop: Travelled for **t** seconds

3rd drop: Travelled for **2t** seconds

2nd drop: Travelled for **3t** seconds

1st drop: Travelled for **4t** seconds (This one is at the ground)

Step 2: Find the total time for the 1st drop Using the equation: **$H = (1/2) \times g \times T^2$**

$16 = (1/2) \times 10 \times (4t)^2$

$16 = 5 \times 16 \times t^2$

$1 = 5 \times t^2$

$t^2 = 1/5$

Step 3: Calculate the distance of the 2nd drop from the top The 2nd drop has been falling for a time of **3t**.

Distance from top (s) = $(1/2) \times g \times (3t)^2$

$s = (1/2) \times 10 \times 9 \times t^2$

Substitute $t^2 = 1/5$: $s = 5 \times 9 \times (1/5)$

s = 9 meters

Final Result: The second drop is **9 meters** away from the top. The correct option is **(2).**

24) **Question Explanation :** Match the different segment of a displacement-time (s-t) graph with the corresponding characteristics of velocity.

Concept: Slope of Displacement-time graph.

Solution: Slope of the displacement-time graph gives velocity

v = slope of s vs t graph

$$v = \frac{ds}{dt}$$

Lets analyze each segment of the graph

(i) Segment OA: The curve is parabolic and the slope is increasing. It means the velocity is increasing with time.

(ii) Segment AB: The segment is a straight line with a constant positive slope.

 Velocity is constant and independent of time.

(iii) Segment BC: The curve is parabolic but the slope is decreasing.

It means Velocity is decreasing with time.

(iv) Segment CD: The line is parallel to the time axis, so the slope is zero
 Velocity is zero.

Final Answer: Option (3)

25) **Asking about :**
Find the velocity when acceleration is zero.

Concept :
Velocity is the first derivative of displacement, and acceleration is the derivative of velocity.

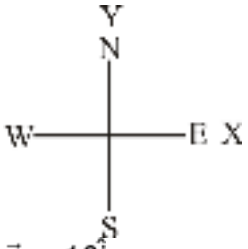
Formula :
Velocity: $v = \frac{dS}{dt}$
Acceleration: $a = \frac{dv}{dt}$

Explanation/Calculation :
Given: $S = t^3 - 6t^2 + 3t + 4$
1. Find velocity:
$$v = \frac{dS}{dt} = 3t^2 - 12t + 3$$
2. Find acceleration:
$$a = \frac{dv}{dt} = 6t - 12$$
3. Set acceleration to zero:
$6t - 12 = 0 \Rightarrow t = 2$
4. Find velocity at t =2:
$v = 3(2)^2 - 12(2) + 3$
$= 3(4) - 24 + 3 = 12 - 24 + 3 = -9 ms^{-1}$
Final Answer :
$v = -9 ms^{-1}$

Hence, the correct answer is option (4).

26) **Question Explanation:** To find the net displacement of body.
Concept: Distance and displacement.
Formula: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ & $r = \sqrt{x^2 + y^2 + z^2}$
Solution:
$\vec{A} = 6\hat{j}$
$\vec{B} = 8\hat{i}$
$\vec{C} = 10\hat{k}$
Net displacement $\vec{S} = \vec{A} + \vec{B} + \vec{C}$
$\vec{S} = 6\hat{j} + 8\hat{i} + 10\hat{k}$
or $|\vec{S}| = \sqrt{(8)^2 + (6)^2 + (10)^2}$
S = $10\sqrt{2}$m
Final Answer : (1)

27) **Question Explanation :** Body is moving with velocity 10 m/s towards north and after 2 sec. Its velocity becomes 20 m/s towards west. Find average acceleration of body.
Concept: Kinematics - average acceleration.
Formula : $\vec{a}_{av} = \dfrac{\vec{v}_f - \vec{v}_i}{\Delta t}$
Solution:



$\vec{v}_i = 10\hat{j}$
$\vec{v}_f = -20\hat{i}$
$\vec{a}_{av} = \dfrac{\vec{v}_f - \vec{v}_i}{\Delta t}$
$= \dfrac{-20\hat{i} - 10\hat{j}}{2}$
$|\vec{a}_{av}| = \dfrac{\sqrt{20^2 + 10^2}}{2}$
$= 5\sqrt{5}\ m/s^2$
Final answer : option (1)

28) **Explain question :** Velocity depends on the position, then acceleration at the given position will be.
Concept: This question is based on motion with variable acceleration.
Solution :
$v = 3x^2 - 2x$
$\dfrac{vdv}{dx} = a = \left(3x^2 - 2x\right)(6x - 2)$
$a(x = 2) = (3(2)^2 - 2 \times 2)(6 \times 2 - 2)$
$= (3 \times 4 - 4)(12 - 2)$
$= 8 \times 10 = 80\ m/s^2$

Final answer : 80 m/s^2

29) **Explaining :** R = 30 M, h_1 = 150m, h_2 = 27.5m to
ask : speed with which a ball must be thrown
Concept : This question is based on horizontal projectile motion
Solution :
h = 150 – 27.5 = 122.5 m

$$t = \sqrt{\frac{2 \times 122.5}{9.8}} = 5 \text{ sec.}$$

Hence, $u = \frac{R}{t} = \frac{30}{5} = 6 \text{ m s}^{-1}$
Final Answer : (2)

30) **Explaining :** Four persons P, Q, R and S of same mass travel with same speed u along a square of side 'd' such that each one always faces the other. After what time will they meet each other ?
Concept : This question is based on Relative motion
Solution :

$$t = \frac{d}{u + u\cos 90} = \frac{d}{u}$$

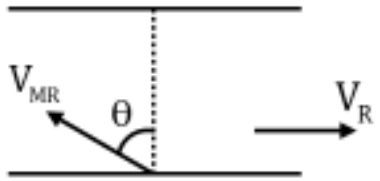
Final Answer : (1)

31) **Explaining :** d = 150 m, v_T = 10 m/s, v_P = 5m/s, find time taken to cross the train
Concept : This question is based on Relative motion in 1D
Solution :

$$t = \frac{d_{rel}}{v_{rel}} = \frac{150}{10 + 5} = 10 \text{ s}$$

Final Answer : (4)

32) **Explaining :** A man can swim in still water with a speed 2 times that of the river then at what angle with downstream he must swim to cross the river along shortest possible path :-
Concept : This question is based on River - man problem
Solution :



$$\sin\theta = \frac{V_R}{V_M} = \frac{v}{2v}$$

θ = 30°
angle from down stream = 90° + 30° = 120° **Final Answer : (2)**

33) **Explaining :** An aeroplane moving horizontally at 720 km/hr drops a food packet from a height 396.9m.
ask : Find the time to reach the ground and horizontal range
Concept : This question is based on oblique projectile motion

Solution :

$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 396.9}{9.8}} = 9$ sec

and u = 720 km/hr = 200 m/s

 R = u × t = 200 × 9 = 1800 m.

Final Answer : (4)

34) **Explaining :** A body moves along the circular path as shown in figure. Find magnitude of displacement between point A & B

Concept : This question is based on Distance and displacement

Solution :

$s = 2R \sin\left(\frac{\theta}{2}\right)$ $s = 2R \sin\left(\frac{60}{2}\right)$

$s = 2R\frac{1}{2}$

s = R

Final Answer : (3)

35) **Explaining :** The equation of motion of a projectile is: $y = 12x - \frac{5}{9}x^2$. The horizontal component of velocity is 3 ms^{-1}. Given that g = 10 ms^{-2}, what is the range of the projectile?

Concept : This question is based on equation of trajectory

Solution :

At x = R (range), y = 0

So, $12x - \frac{5x^2}{9} = 0$

⇒ x = 0, x = 21.6 m

OR compare given equation with

$y = x\tan\theta\left(1 - \frac{x}{R}\right)$

with $y = 12x - \frac{5x^2}{9}$

or $y = 12x\left(1 - \frac{5x}{9 \times 12}\right)$

$R = \frac{9 \times 12}{5} = 21.6$ m

Final Answer : (2)

36) $\vec{v} = a\hat{i} + (b - ct)\hat{j}$

$v_x = a$; $v_y = (b - ct)$

 At maximum height

$v_y = 0$

$b - ct = 0$

$t = \frac{b}{c}$

So time of Flight $T = \frac{2b}{c}$

Range R = $(v_x)(T)$

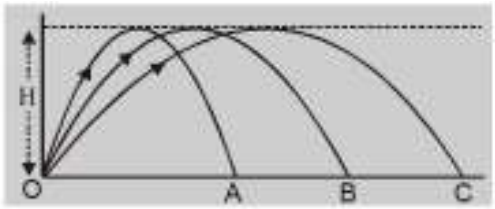
$$\boxed{R = \frac{2ab}{c}}$$

37) **Question Explanation:**

A. **Observation:** The figure shows three projectile trajectories (A, B, and C) starting from the same point.
B. **Key Visual Data:** All three projectiles reach the same maximum height (H_{max}) but they have different horizontal ranges ($R_C > R_B > R_A$)

Concept: Projectile Motion: Maximum Height and Time of Flight.

Solution:



$H = \frac{u^2 \sin^2\theta}{2g}$ which is same for all

$\Rightarrow$ u sinθ is same for all

$T = \frac{2u \sin\theta}{g}$ {same for all}

for large θ value of sinθ is large and hence u will be less

$\Rightarrow u_A < u_B < u_C$

i.e. launch speed is maximum for particle C horizontal range $R = u_x \times T$

T is same for all and R_C is maximum for C

$\Rightarrow u_x$ is maximum for C

Final Answer : Option (3)

38) **Explanation :-** Find maximum height achieved.

Concept & formula :- $H = \frac{4y^2}{2g}$

Solution :

Max height can be compared according to vertical component of velocity

$(V_A)_1 = 2\ V \sin 30° = V$; $(V_B)^0 = V \sin 60° \quad V \times \frac{\sqrt{3}}{2}$

So A achieve more height

Final Answer : (1)

39) **Explaining :** This question is based on Unit conversion.

Concept :

$1Å = 10^{-10}$ m

1 fermi = 10^{-15} m

1 light year = 9.46×10^{15} m

Solution :
Astronomical unit is a unit of length. It is a mean distance between sun and earth.
1 Astronomical unit (1 AU) = 1.496×10^{11} m
Final Answer : (4)

40) **Explaining :** A dimensionally incorrect equation is always incorrect.
Concept : This question is based on rules of dimension.
Solution :
Dimensionally correct equation may be correct or incorrect but dimensionally incorrect equation is always incorrect e.g. $p = \frac{mv}{t}$ is dimensionally as well as experimentally wrong.
Final Answer : (3)

41) **Explaining :** Dimensions in L.H.S. = Dimensions in R.H.S.
Concept : This question is based on relation between different physical quantities.
Solution :
Let $[M] \propto [F^aL^bT^c]$
So, using dimensions, we have
$[ML^0T^0] = k[MLT^{-2}]^a[L]^b[T]^c$
$\Rightarrow$ $a = 1, a + b = 0 \Rightarrow b = -1$
and $-2a + c = 0 \Rightarrow c = 2$
So, unit of mass is $[FL^{-1}T^2]$
Final Answer : (3)

42) **Explaining :** $n_1u_1 = n_2u_2$
Concept : This question is based on conversion between system of units.
Solution :
Unit of $F = [M^1L^1T^{-2}]$
$$= \left[\frac{10g}{1000g}\right]^1\left[\frac{10cm}{100cm}\right]^1\left[\frac{0.1\,sec}{1\,sec}\right]^{-2}$$
$F = 0.1$ N
Final Answer : (1)

43) **Explaining :** Least count = 1 MSD – 1 VSD
Concept : This question is based on Least count of vernier callipers
Solution :
Least count of main scale = $\frac{1cm}{10} = 0.1cm$
Least count of v.s.
Value of 10 parts of V.S.
= Value of 8 parts on M.s.
10 parts = 8 [0.1 cm]
1 part = 0.08 cm
Least count = Value of 1 part of M.S. - Value of 1 part of V.S.
L.C. = 0.10 cm – 0.08 cm = 0.02 cm
Final Answer : (2)

44) **Explaining :** Mean (Average) absolute error = $\frac{|\Delta a_1| + |\Delta a_2| + \ldots\ldots\ldots |\Delta a_n|}{n}$
Concept : This question is based on to find average absolute error.
Solution :
Average value
$= \frac{2.63 + 2.56 + 2.42 + 2.71 + 2.80}{5} = 2.62s$
Now,
$|\Delta T_1| = 2.63 - 2.62 = 0.01$
$|\Delta T_2| = 2.62 - 2.56 = 0.06$
$|\Delta T_3| = 2.62 - 2.42 = 0.20$
$|\Delta T_4| = 2.71 - 2.62 = 0.09$
$|\Delta T_5| = 2.80 - 2.62 = 0.18$
$\Delta T = \frac{|\Delta T_1| + |\Delta T_2| + |\Delta T_3| + |\Delta T_4| + |\Delta T_5|}{5}$
$= \frac{0.54}{5} = 0.108$
$\Rightarrow \Delta T \simeq 0.11s$
Final Answer : (2)

45) **Question Explanation:**
Summing with decimal precision.

Concept: Significant Figures (Addition).

Solution:
In addition, the final result must be rounded to the same number of decimal places as the measurement with the least decimal places.
Sum: 2.331 cm + 2.1 cm = 4.431 cm.
• Precision: 2.1 has the least number of decimal places (one).

• Rounding: Rounding 4.431 to one decimal place gives 4.4.

Final Answer: Option (3)

CHEMISTRY

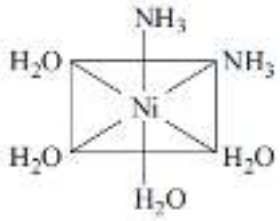
46) **Explaining :** Isomerism.
Concept : This question is based on isomerism.
Solution : A → fac− ; B → mer −
Final Answer : (4)

47) **Explaining :** IUPAC Nomenclature.
Concept : This question is based on IUPAC Nomenclature.
Solution : As per Rule, the IUPAC name of the complex
is tetraamminedichlorocobalt(III)chloride.
Final Answer : (1)

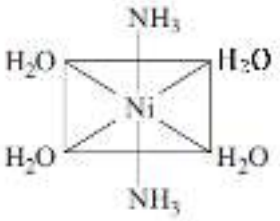
48) **Explaining :** Werner’s theory

Concept : This question is based on Werner's theory.
Solution :
The primary and secondary valencies of chromium in the complex ion, dichlorodioxalato chromate(III) are 3 and 6 respectively.
$[Cr(Cl)_2(C_2O_4)_2]^{-3}$ is complex.
Final Answer : (4)

49) **Explaining :** Isomerism.
Concept : This question is based on isomerism.
Solution :
(II) and (III) show only geometrical isomerism.
$[Ni(H_2O)_4(NH_3)_2]^{2+}$ show cis and trans-isomers. These are followings

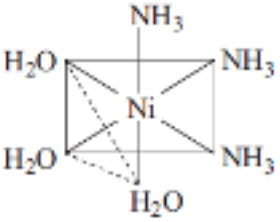


cis

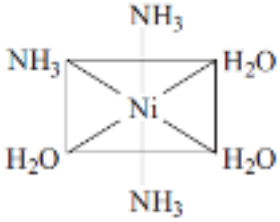


trans

$[Ni(H_2O)_3(NH_3)_3]^{2+}$ show facial and meridional geometrical isomerism.



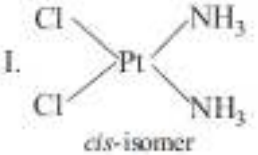
fac-isomer



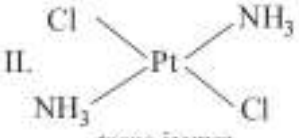
mer-isomer

Final Answer : (3)

50) **Explaining :** Geometrical isomerism.
Concept : This question is based on Geometrical isomerism.
Solution :



cis-isomer

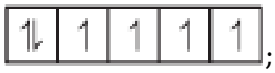
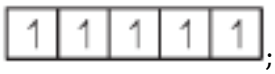
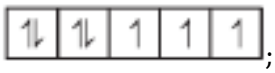


trans-isomer

Final Answer : (1)

51) **Explaining :** Magnetic moment
Concept : This question is based on magnetic moment.
Solution :
The complexes, in which metals ions have same number of unpaired electrons will have same magnetic moment.

Complex ion	Electronic configuration of metal ion	Number of unpaired electrons (n)
$[Cr(H_2O)_6]^{2+}$	Cr^{2+}; $[Ar]3d^4$	1 1 1 1 ; 4

Complex ion	Electronic configuration of metal ion	Number of unpaired electrons (n)
$[Fe(H_2O)_6]^{2+}$	Fe^{2+}; $[Ar]3d^6$	 ; 4
$[Mn(H_2O)_6]^{2+}$	Mn^{2+}; $[Ar]3d^5$	 ; 5
$[CoCl_4]^{2-}$	Co^{2+}; $[Ar]3d^7$	 ; 3

Final Answer : (1)

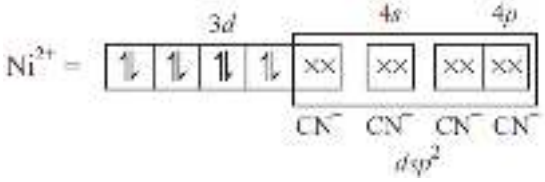
52) **Explaining :** Salts
Concept : This question is based on salts.
Solution : Mohr's salt gives NH_4^+, Fe^{2+} and SO_4^{2-} ions in aqueous solution because it is a double salt.
Final Answer : (1)

53)

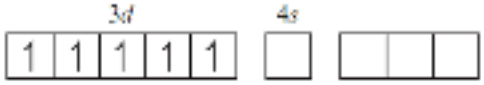
Explaining : Geometry, hybridisation and magnetic moment.
Concept : This question is based on geometry, hybridisation and magnetic moment.
Solution :
For $[Ni(CN)_4]^{2-}$, oxidation state of Ni is +2.
CN^- = strong field ligand

Ni^{2+} (ground state) = 

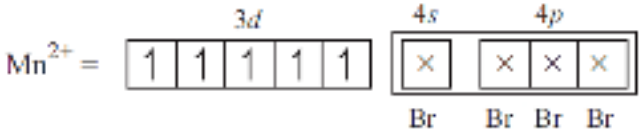
In $[Ni(CN)_4]^{2-}$



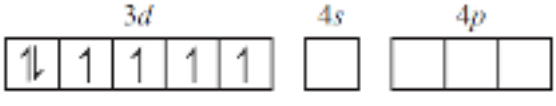
dsp^2 hybridisation, i.e. square planar geometry, zero unpaired electron, i.e. zero magnetic moment.
For $[MnBr_4]^{2-}$, oxidation state of Mn is +2.
Br^- = weak field ligand
Mn^{2+} (in ground state)



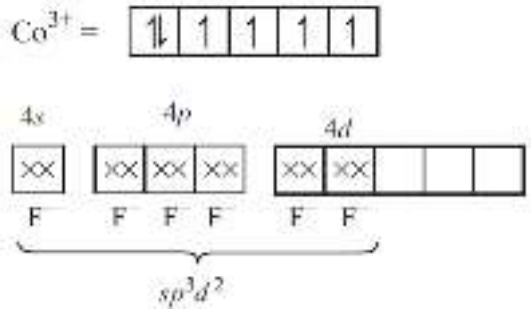
In $[MnBr_4]^{2-}$,



sp^3 hybridisation, i.e, tetrahedral geometry, five unpaired electrons, i.e. magnetic moment = 5.9 as [$\mu = \sqrt{n(n+2)}$ and n = 5]

Co^{3+} in ground state = 

For $[CoF_6]^{3-}$, oxidation state of Co is +3
F^- = weak field ligand
In $[CoF_6]^{3-}$



sp^3d^2 hybridisation, i.e. octahedral geometry four unpaired electrons, i.e. magnetic moment is 4.91 BM.
Final Answer : (4)

54) **Explaining :** Werner's theory.
Concept : This question is based on Werner's theory.
Solution : Secondary valency responsible for geometry.
Final Answer : (2)

55) **Explaining :** Spectrochemical series.
Concept : This question is based on spectrochemical series.
Solution :
Based on spectrochemical series, ligands arranged in increasing order of crystal field strength are as
$NH_3 < en < CN^- < CO$
Final Answer : (2)

56) **Explaining :** Synergic bonding.
Concept : This question is based on synergic bonding.
Solution : Metal-carbon bond in metal carbonyls possess both σ and π-character.
Final Answer : (3)

57) **Explaining :** The order of energy absorbed which is responsible for the color of complexes
(A) $[Ni(H_2O)_2(en)_2]^{2+}$
(B) $[Ni(H_2O)_4(en)]^{2+}$ and
(C) $[Ni(en)_3]^{2+}$
is
Concept : This question is based on energy absorbed by the complex.
Solution :
Stronger the field strength of ligand, higher will be the energy absorbed by the complex.
⇒ 'en' has a stronger field strength than 'H_2O' according to spectrochemical series
 Correct order of energy absorbed will be:
$[Ni(en)_3]^{2+} > [Ni(H_2O)_2(en)_2]^{2+} > [Ni(H_2O)_4(en)]^{2+}$
i.e. (C) > (A) > (B)
Final Answer : (3)

58) **Explaining :** Increasing order of EAN of the metal in $[Ni(CN)_4]^{2-}$, $[Fe(CN)_6^{4-}]$ and $[Cu(CN)_4]^{3-}$

Concept : This question is based on EAN of the metal.

Solution :

Complex	z	OS of metal	Electron gained due to ligands	EAN
$(Ni(CN)_4)^{2-}$	28	+2	8	28–2+8 = 34
$(Fe(CN)_6^{4-}$	26	+2	12	26–2+12 = 36
$(Cu(CN)_4)^{3-}$	29	+1	8	29–1+8 = 36

Final Answer : (2)

59) **Explaining :** Which of the following are paramagnetic ?

A. $[Ni(Cl)_4]^{2-}$ B. $Ni(CO)_4$ C. $[Ni(CN)_4]^{2-}$ D. $[Ni(H_2O)_6]^{2+}$ E. $Ni(PPh_3)_4$

Choose the correct answer from the options given below:

Concept : This question is based on magnetic behaviour.

Solution : -

A. $[NiCl_4]^{2-}$

Nickel oxidation state : Ni^{2+}

Electron configuration : $3d^8$

Hybridization : sp^3

Unpaired electrons : 2

Magnetic property : Paramagnetic

B. $Ni(CO)_4$

Nickel oxidation state : Ni (neutral)

Electron configuration : $3d^8\ 4s^2 \approx 3d^{10}4s^0$

Hybridization : sp^3

Unpaired electrons : 0

Magnetic property : Diamagnetic

C. $[Ni(CN)_4]^{2-}$

Nickel oxidation state : Ni^{2+}

Electron configuration : $3d^8$

Hybridization : dsp^2

Unpaired electrons : 0

Magnetic property : Diamagnetic

D. $[Ni(H_2O)_6]^{2+}$

Nickel oxidation state : Ni^{2+}

Electron configuration : $3d^8$

Hybridization : sp^3d^2

Unpaired electrons : 2

Magnetic property : Paramagnetic

E. $Ni(PPh_3)_4$

Nickel oxidation state : Ni (neutral)

Electron configuration : $3d^84s^2 \approx 3d^{10}4s^0$

Hybridization : sp^3

Unpaired electrons : 0

Magnetic property : Diamagnetic

Overall, complexes A and D are paramagnetic as they have unpaired electrons.

Final Answer : (1)

60) **Explaining :** Identify the homoleptic complexes that is low spin.
Concept : This question is based on homoleptic complexes.
Solution :
(A) $[Fe(CN)_5NO]^{-2}$ → Heteroleptic, Fe^{+2}, $3d^6$, $t_{2g}^6 e_g^0$, d^2sp^3, Low spin (3d series + SFL)
(B) $[CoF_6]^{-3}$ → Homoleptic, sp^3d^2, High spin, Co^{+3}, $3d^6$ (3d series + WFL)
(C) $[Fe(CN)_6]^{-4}$ → Homoleptic
Fe^{+2}, $3d^6$, d^2sp^3, $t_{2g}^6 e_g^0$ Low spin
(3d series + SFL)
(D) $[Co(NH_3)_6]^{+3}$ → Homoleptic, Co^{+3} $3d^6$, d^2sp^3, $t_{2g}^6 e_g^0$, Low spin (3d series + SFL)
(E) $[Cr(H_2O)_6]^{+2}$ → Homoleptic
Cr^{+2} $3d^4$, d^2sp^3, High spin $t_{2g}^3 e_g^1$
(3d series + WFL)
Final Answer : (3)

61) **Explaining :** Which complex is most stable ?
Where K_d = Dissociation constant
Concept : This question is based on stability of complex.
Solution : [$K_d \downarrow \Rightarrow$ Stability of complex$\uparrow$]
Final Answer : (3)

62) **Explaining :** Which is inner orbital and low spin complex ?
Concept : This question is based on VBT.
Solution :

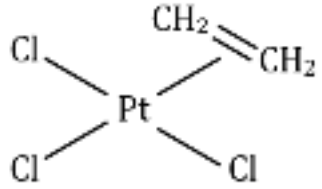
$[NiCl_4]^{-2}$	$[Ni(CN)_4]^{2-}$
Ni^{+2} : $3d^8$	Ni^{2+} : $3d^8$
Cl^- : WFL	CN^- : SFL
hyd :sp^3	hyd : dsp^2
spin : high	spin : low
outer orbital complex	inner orbital complex

$[CoF_6]^{3-}$	$[FeF_6]^{3-}$
Co^{3+} : $3d^6$	Fe^{3+} : $3d^5$
F^- : WFL	F^- : WFL
hyd : sp^3d^2	hyd : sp^3d^2
spin : high	spin : high
outer orbital complex	outer orbital complex

Final Answer : (2)

63) **Explaining :** Bond length C = C in Ziese's salt.
Concept : This question is based on bond length of C = C.

Solution :

Ziese's salt : 

The C—C bond length in salt is longer than the isolated ethene. This increase in length is due to the donation of electrons from (C=C) π bond to the Pt.

Final Answer : (1)

64)

Explaining : From the given complex identify the given complex show geometrical isomerism.
Concept : This question is based on Isomerism in coordination compound.
Solution :

$[Cr(H_2O)_4Cl_2]^+$ – $M_{a_4b_2}$ – Show GI

$[Pt(NH_3)_3Cl]^-$ – M_{a_4} – No GI

$[Co(NH_3)_6]^{+3}$ – M_{a_6} – No GI

$[Co(CN)_5(NC)]^{-3}$ – M_{a_5b} – No GI

Final Answer : (1)

65)

Question Explanation: Which complex have least conductivity or least no. of ions.
Concept: This question is based on Werner's theory.
Solution:
if no. of ions are higher conductivity is higher

(1) $\left[Co(NH_3)_4Cl_2\right]Cl \rightarrow \left[Co(NH_4)_4Cl_2\right]^+ + Cl^-$ → 2- ions

(2) $\left[Co(NH_3)_4Cl_2\right] \rightarrow \left[Co(NH_3)_2Cl_2\right]$ ⇒ 0 ions

(3) $\left[Co(NH_3)_5Cl\right]Cl_2 \rightarrow \left[Co(NH_3)_5Cl\right]^{+2} + 2Cl$ ⇒ 3 ions

(4) $\left[Co(NH_3)_6\right]Cl_3 \rightarrow \left[Co(NH_3)_6Cl\right]^{+3} + 3Cl$ ⇒ 4 ions

Hence, least conductivity by option (2)
Final Answer: Option (2)

66)

Explaining : Oxidation state of Fe in prussian blue.
Concept : This question is based on Introduction to coordination compound.
Solution : Prussian blue – $4Fe_4[Fe(CN)_6]_3 \rightarrow 4Fe^{+3} + 3[\underline{Fe}(CN)_6]^{4-}$

(III) x

x – 6 = –4

x = +2

Final Answer : (2)

67)

Explain Question: Asking about electronic configuration of d^5 in octahedral weak-field complex.

Concept: If pairing energy (P) > Δ_0, high-spin configuration occurs; electrons occupy orbitals to maximize unpaired electrons.
Solution: d^5 octahedral, weak-field ligand → high-spin → maximize unpaired electrons.
Example : $[Mn(H_2O)_6]^{2+}$ has d^5 configuration and weak-field H_2O ligands → high-spin $t_{2g}^3e_g^2$.
Final answer: Option (2)

68) **Explaining :** From the following ligand which will not act as chelating ligand.
Concept : This question is based on Types of ligands.
Solution :
Bi or Polydentate ligand act as chelating ligand.
N_2H_4 – monodentate ligand.
$NH_2–CH_2–CH_2–NH_2$ – Bidentate ligand.
$EDTA^{-4}$ – Hexadentate ligand.
dmg – Bidentate ligand.
Final Answer : (1)

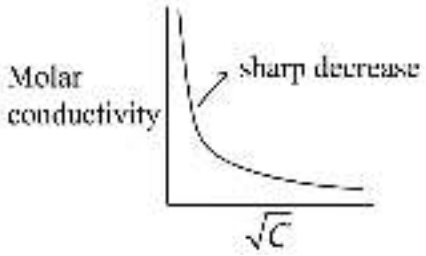
69) **Explaining :** How can an electrochemical cell be converted into an electrolytic cell ?
Concept : This question is based on electrochemical and electrolytic cell
Solution :
An electrochemical cell can be converted into an electrolytic cell by applying an external potential greater than E°cell. The entire cell reaction will be reversed. The oxidised species will be reduced and the reduced species will be oxidised.
Final Answer : (1)

70)

Explaining : What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M $ZnSO_4$?
Concept : This question is based on oxidising and reducing power
Solution :
In the reaction of copper with zinc ions, zinc is more active (easily oxidised) than copper; thus, no reaction will take place.
Final Answer : (4)

71)

Explaining : The molar conductivity of a weak electrolyte when plotted against the square root of its concentration, which of the following is expected to be observed?
Concept : This question is based on effect of dilution on Molar conductivity for weak electrolytes
Solution :
For a weak electrolyte molar conductivity decreases sharply with increase in concentration.



Correct
From the graph it is clear that, molar conductivity decreases sharply with increase in concentration.
Final Answer : (2)

72) **Explaining :** The amount of the aluminium deposited at the cathode is?
Concept : This question is based on Faraday's Law of electrolysis.
Solution :
$Al^{3+} + 3e^- \rightarrow Al$

$$\text{Moles of } e^- = \frac{It}{F} = \frac{2 \times 30 \times 60}{96500}$$

Mass of Al deposited

$$= \frac{1}{3} \times \frac{2 \times 30 \times 60}{96500} \times 27 = 0.336 \text{ g}$$

Final Answer : (3)

73) **Explaining :** Predict the one with the strongest oxidising capacity
Concept : This question is based on oxidising power
Solution :
A higher standard reduction potential indicates a stronger oxidizing capacity. A substance with a higher standard reduction potential is more likely to gain electrons and act as an oxidizing agent in a redox reaction. The more positive the reduction potential, the stronger the oxidizing power of a species. From the given options, the higher standard reduction potential is for lead.
Final Answer : (3)

74) **Explaining :** Calculate molar conductance of 0.01M KCI solution.
Concept : This question is based on molar conductance
Solution :

$$\kappa = G \times G^* \quad \left(G = \frac{1}{R}\right)$$

$$R = \frac{1}{10^{-4}} \times G^*$$

$$100 = \frac{1}{10^{-4}} \times G^*$$

$\Rightarrow G^* = 0.01 \text{ cm}^{-1}$

$$50 = \frac{1}{\kappa} \times 0.01$$

$$\kappa = \frac{1}{5} \times 10^{-3}$$

$$\Lambda = \frac{1}{5} \times 10^{-3} \times \frac{1000}{0.01} = 20$$

Final Answer : (2)

75) **Explaining :** From the following select the one which is **not** an example of corrosion.
Concept : This question is based on corrosion.
Solution : Production of hydrogen by electrolysis of water is not an example of corrosion.
Final Answer : (2)

76) **Explaining :** Evaluate the given statements.
Concept : This question is based on application of electrochemical series.
Solution :
In an electrolytic cell with inert electrodes, reduction occurs at the cathode. The ion with the highest (most positive) reduction potential will be reduced (or deposited) first. Therefore, as we increase the applied voltage, the metals will begin to deposit in the order of their reduction potentials.
The order of reduction (deposition) will be:
First, silver (+0.80 V)
Next, mercury (+0.79 V)
Then, copper (+0.24 V)
So, Statement (I) is correct.
For magnesium, the reduction potential is very negative (–2.37 V). In aqueous solution, before reaching such a negative potential, water will be reduced. The cathodic reaction for water typically is:
$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$
which leads to the evolution of hydrogen gas, not oxygen. (Oxygen is produced at the anode during the oxidation of water.)
Thus, Statement (II) is incorrect.
Final Answer : (3)

77) **Explaining :** The number of correct statements from the following is _____.
Concept : This question is based on effect of dilution on conductivity.
Solution :
(A) As the concentration of ions decreases, the conductivity of the solution also decreases, regardless of whether the electrolyte is strong or weak.
(B) The number of ions per unit volume that carry current in a solution increases on dilution. This statement is false. The number of ions per unit volume in a solution decreases on dilution, because the total number of ions in the solution remains the same, but the volume increases.
(C) As the concentration of the electrolyte decreases, the molar conductivity increases because the ions become more separated from each other, and the electrostatic interactions between them become weaker, allowing them to move more freely in the solution.
(D) The variation in molar conductivity with concentration is different for strong and weak electrolytes. Strong electrolytes dissociate completely in solution, so their molar conductivity increases rapidly with dilution, while weak electrolytes dissociate only partially, so their molar conductivity increases more slowly with dilution.
(E) For weak electrolytes, the change in molar conductivity with dilution is due to decrease in degree of dissociation. This statement is false. For weak electrolytes, the change in molar conductivity with dilution is due to an increase in the degree of dissociation, not a decrease.
Final Answer : (2)

78)

Explaining : Find product of fuel cell

Concept : This question is based on fuel cell
Solution : In typical fuel cell
Reactants = H_2, O_2
Products = H_2O
Final Answer : (2)

79) **Explaining :** $\Lambda^0_{m\ (NH_4OH)}$ is equal to ____.
Concept : This question is based on Kohlrausch's law.
Solution : $\Lambda^0_{m\ (NH_4OH)} = \lambda^0_{m(NH_4Cl)} + \lambda^0_{m(NaOH)} - \lambda^0_{m(NaCl)}$
Final Answer : (2)

80) **Explaining :** Match the terms given in Column I with the units given in Column II.
Concept : This question is based on units of electrochemical terms.
Solution :
(i) $\Lambda_m \rightarrow$ (c) S cm^2 mol^{-1}
(ii) $E_{Cell} \rightarrow$ (d) V
(iii) $\kappa \rightarrow$ (a) Scm^{-1}
(iv) G* $\rightarrow$ (b) m^{-1}
Final Answer : (3)

81) **Explaining :** Evaluate the given statements
Concept : This question is based on oxidising power
Solution :
Higher the SRP, more the oxidising power. Therefore, oxidising power of ions is $C^{3+} > A^+ > B^{2+}$.
Final Answer : (1)

82) **Explaining :** Which of the above relations are correct?
Concept : This question is based on emf of an electrochemical cell.
Solution : Theory based
Final Answer : (4)

83) **Explaining :** The oxidation potential of given electrode would be?
Concept : This question is based on standard hydrogen electrode.
Solution :
$H_2 \rightarrow 2H^+ + 2e^-$
1 atm 10^{-10}
$$E_{H_2/H^+} = 0 - \frac{0.059}{2}\log_{10}\frac{(10^{-10})^2}{1}$$
$$E_{H_2/H^+} = 0.59\ V$$
Final Answer : (1)

84) **Explaining :** Find equilibrium constant of the given cell reaction.
Concept : This question is based on relation between equilibrium constant and standard electrode potential.

Solution :
For a cell reaction in equilibrium at 298 K.

$$E^{o}_{cell} = \frac{0.0591}{n} \times \log_{10} K_C$$

$$0.46 = \frac{0.0591}{2} \times \log_{10} K_C$$

$$\log_{10} K_C = \frac{2 \times 0.46}{0.0591} = 15.57$$

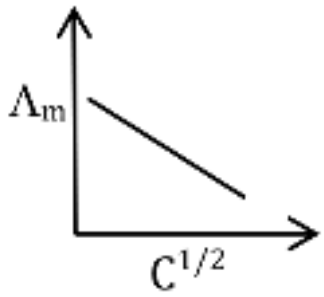
$$K_C = 3.7 \times 10^{15} \simeq 4 \times 10^{15}$$

Final Answer : (4)

85) **Explaining :** Negative E^0 shows the tendency of a species to lose electrons easily compared to H^+/H^2.
Concept : This question is based on If $E^0 < 0 \rightarrow$ stronger reducing agent than H^+/H_2.
If $E^0 > 0 \rightarrow$ stronger oxidising agent than H^+/H_2.
Solution :
Since E^0 is negative, the species donates electrons more easily than hydrogen and acts as a strong reducing
agent.
Final answer : (1)

86)

Explain Question: We need to find suitable electrolyte for given graph.
Concept: This question is based on Debye – Huckel – Onsager equation.
Solution:
$\Lambda_m = \Lambda_m^{\infty} - b\sqrt{c}$:- Only for strong electrolyte



$\Rightarrow$ KCl :- strong electrolyte.
Final answer: (1)

87)

Explanation the Question: The question asks to calculate the cell EMF (E_{cell}) for the given hydrogen concentration cell.
Concept: This question is based on Nernst Equation for a Concentration Cell
Solution:
Overall Reaction (n=2): $2H^+$(aq. 0.1M) $\rightarrow 2H^+$(aq. 0.001M)

$$E_{cell} = E^0_{cell} - \frac{0.0591}{n} \log Q$$

$$Q = \frac{[H^+]^2_{anode}}{[H^+]^2_{cathode}}$$

$$Q = \frac{(0.001)^2}{(0.1)^2} = \frac{(10^{-3})^2}{(10^{-1})^2} = \frac{10^{-6}}{10^{-2}} = 10^{-4}$$

$$E_{cell} = 0 - \frac{0.0591}{2}\log(10^{-4})$$

$$E_{cell} = -\frac{0.0591}{2} \times (-4)$$

$E_{cell} = -(0.02955) \times (-4)$

$E_{cell} = +0.1182$ V

Final Answer : (1)

88) **Explaining :** A device that converts energy of combustion of fuel like hydrogen and methane directly into electrical energy is known as :-

Concept : This question is based on Fuel cell.

Solution : Fuel cell

Final Answer : (4)

89) **Explaining :** Which of the following expression correctly represents the equivalent conductance at infinite dilution of $Ca_3(PO_4)_2$ Given $\Lambda^o_{Ca^{+2}}$ and $\Lambda^o_{PO_4^{-3}}$ are the equivalent conductance at infinite dilution of the respective ions :-

Concept : This question is based on application of Kohlrausch's Law.

Solution :

$Ca_3(PO_4)_2 \rightarrow 3Ca^{2+} + 2PO_4^{3-}$

$\Lambda^o_{eq} = \Lambda^o_{Ca^{2+}} + \Lambda^o_{PO_4^{3-}}$

Final Answer : (1)

90)

Explaining : A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode ?

Concept : This question is based on Faraday's Law of electrolysis.

Solution :

$Q = I \times t$

$= 1.5 \times 10 \times 60$

$= 900C$

Reaction occuring at the cathode is :

$Cu^{+2} + 2e^- \rightarrow Cu$

$2F = 2 \times 96500$ deposit 1 mole Cu(63.5g)

$\Rightarrow$ 900 C will deposit = 0.296 g

Final Answer : (4)

BIOLOGY

91) **Solution :** All the given statements are true.

Final Answer : (1)

92) **Solution :** Hilum represents the junction between ovule and funicle.
Final Answer : (3)

93) **Solution :** Stigma is the receptive surface for pollen grains. When more than one carpel is fused it is referred to as syncarpous conditions.
Final Answer : (3)

94) **Solution :** In mango one ovule is present in the ovary.
Final Answer : (3)

95) **Solution :** Pollen grain (n) = 28
Nucellus (2n) = 56
Final Answer : (3)

96) **Solution :** Pollen grain represents male gametophyte.
Final Answer : (4)

97) **Solution :** Polar nuclei, pollen grain, embryo sac, synergids, microspore and megaspore are haploid in nature.
Final Answer : (1)

98) **Solution :** *Amorphophallus* flower provide floral rewards to the pollinator in the form of providing safe place to lay eggs.
Final Answer : (2)

99) **Solution :** Water pollination in freshwater *Vallisneria.*
Final Answer : (4)

100) **Solution :** Pollination by water is rare in flowering plants and is limited to 30 genera, mostly monocotyledons.
Final Answer : (4)

101) **Solution :** Wind pollinated flowers have usually a single ovule in each ovary.
Final Answer : (4)

102) **Solution :** These are common pollinating agents in angiosperms.
Final Answer : (4)

103) **Solution :**
Double fertilization = Syngamy + Triple fusion
Triple fusion = Double fertilization – Syngamy
Final Answer : (4)

104) **Solution :** Endosperm is a nutritive tissue and it is triploid in angiosperms because endosperm is formed by fusion of secondary nucleus and a male gamete. It is used by developing embryo.
Final Answer : (1)

105) **Solution :** A : Polar nuclei, B : Male gametes
Final Answer : (1)

106)

Explain Question : The question asks to identify the group of plants in which the nucellus persists in the seed as perisperm.
Concept : Sexual Reproduction in Flowering Plants Seed structure – Perisperm
Solution : In some seeds the nucellus is not completely consumed during development and remains as perisperm. Typical examples of plants having persistent nucellus (perisperm) are black pepper and beet.
Final answer : (1)

107) **Explain Question :** The question asks to identify the labeled parts A, B, and C in the diagram of a dicot embryo.
Concept : Dicot Embryo Structure.
Solution : The figure shows a dicotyledonous embryo:
A points to the embryonic shoot tip/bud, the Plumule.
B points to the large food-storing leaves, the Hypocotyl.
C points to the embryonic axis below the cotyledons, the Radicle
Final Answer : (3)

108) **Solution :** In the development of monocot plants (like grasses), the epicotyl is part of the embryonic shoot that will grow into the plant's stem. It contains the shoot apex (growing tip) and a few leaf primordia (early-stage leaf structures). This is enclosed in a hollow protective structure called the coleoptile.
1. Coleoptile: The coleoptile is protective sheath that surrounds the epicotyl and the shoot apex in monocots, It is a hollow, tubular structure that protects the yound shoot as it emerges for the soil. Once the plant begins to grow, the coleoptile pushes upward and eventually splits open, allowing the shoot to emerge.
Final Answer : (1)

109) **Solution :** Recently excavated seed from arctic region germinated after 10000 year is Lupine.
Final Answer : (1)

110) **Solution :** Apomictic seeds are more advantageous than hybrid seeds due to all of the given reasons.
Final Answer : (4)

111) **Solution :** Fruit production without fertilisation is called parthenocarpy.

Final Answer : (2)

112) **Solution :** The phenomenon of occurrence of more than one embryo in a seed is known as polyembryony.
Final Answer : (3)

113) **Solution :** Polyembryony is absent in China rose.
Final Answer : (4)

114) **Solution :** Unisexual flowers prevent self-pollination.
Final Answer : (3)

115) **Question Explanation:** Sequence of anther wall layers.
Concept: Anther Wall Layers
Solution: The wall of a young anther has four layers from outer to inner: the protective **epidermis**, followed by the **endothecium**, then the **middle layers**, and finally the innermost **tapetum**. The tapetum provides nutrition to the developing pollen grains.
Final Answer: (3)

116) **Solution :** The portion of embryonal axis between plumule (future shoot) and cotyledons is called epicotyl.
Final Answer : (2)

117) **Solution :** In apple and strawberry thalamus also contributes to fruit formation.
Final Answer : (3)

118) **Solution :** (d) Scutellum is the cotyledon found in embryo of grasses.
Final Answer : (3)

119) **Solution :** Ploidy of :
(i) nucellus = 2n
(ii) megaspore mother cell = 2n
(iii) functional megaspore = n
(iv) embryo sac = n
Final Answer : (3)

120) **Solution :** Endosperm is formed due to triple fusion.
Final Answer : (2)

121) **Solution :** The epididymis leads to vas deferens that ascends to the abdomen and loops over the urinary bladder.
Final Answer : (2)

122) **Solution :** Foreskin is a double-layered fold of smooth muscle tissue, blood vessels, neurons, skin, and mucous membrane that covers and protects the glans penis and the urinary meatus when the penis is not erect. The foreskin is mobile, fairly stretchable, and acts as a natural lubricant.
Final Answer : (2)

123) **Solution :** The vasa efferentia leave the testis and open into epididymis located along the posterior surface of each testis. The epididymis leads to vas deferens. Or we can say that the vasa efferentia or efferent ducts connect the rete testis with the initial section of the epididymis.
Final Answer : (3)

124) **Solution :** Ejaculation of human male contains about 200 – 300 million sperms, of which for normal fertility 60 % sperms must have normal shape and size and at least 40 % must show energetic motility.
Final Answer : (3)

125) **Solution :** The zona pellucida is a glycoprotein thick noncellular membrane surrounding the plasma membrane of an oocyte or secondary oocytes.
Final Answer : (4)

126) **Solution :** The milk produced during the initial few days of lactation is called colostrum. This contains several antibodies absolutely essential to develop resistance for the new-born babies.
Final Answer : (4)

127) **Solution :** Graafian follicle releases secondary oocyte from the ovary by the process of ovulation. Primary oocyte within the tertiary follicle grows in size and completes its first meiotic division.
Secondary follicle transforms in to tertiary follicle which is characterised by fluid filled cavity called antrum.
Final Answer : (2)

128) **Solution :** Menstrual cycle is the cycle of natural changes that occurs in the uterus and ovary as an essential part of making sexual reproduction possible. Menstrual fluid cannot easily clot. The end of the cycle of menstruation is called menopause. During the follicular phase, gonadotropins (LH and FSH) increase gradually and stimulate follicular development as well as secretion of estrogen by growing follicles.
Final Answer : (4)

129) **Solution :** The process of fertilization takes place in the ampullary part of fallopian tube. Uterus does not secrete any steroid hormones. It's the ovary which secretes the steroid hormones.
Final Answer : (3)

130) **Solution :** Statement (i) and (iv) are correct. Ovary is the primary female sex organ which produces female gamete (ovum) and steroid hormones. Ovarian stroma is divided into peripheral cortex and inner medulla.
Mammary gland is a paired structure that contains glandular tissue and variable amount of tissue. It is the characteristic of all female mammals.
Final Answer : (1)

131) **Solution :** Fertilization is the fusion of male and female gametes to form zygote. During fertilization only head of the sperm enters egg. After that polyspermy is avoided by fertilization membrane.
Final Answer : (3)

132) **Solution :** In female, Graafian follicle forms corpus luteum after ovulation. The cells of corpus luteum are called luteal cells. The cytoplasm of luteal cells have yellow granules called lutein which secrete the hormone progesterone to maintain pregnancy if fertilization takes place. In the absence of fertilization, corpus luteum degenerates and forms corpus albicans and there is decrease in progesterone level also.
Final Answer : (2)

133) **Solution :** Seminiferous tubules are located in the testes, which are the specific location of meiosis, and the subsequent formation of sperms. Rete testis is a network of small tubules found in the part of the testicle that carries sperm. Leydig cells or interstitial cells of Leydig produce the hormone called testosterone which contributes to male secondary sexual characters. Prepuce is the foreskin, skin surrounding and protecting the head of the penis as well as clitoris.
Final Answer : (2)

134) **Solution :** A spermatogonium divides into two primary spermatocytes. Each primary spermatocyte duplicates its DNA and subsequently undergoes meiosis I to produce two haploid secondary spermatocytes. Each of the two secondary spermatocytes further undergoes meiosis II to produce two haploid spermatids.
Final Answer : (4)

135) **Solution :** A-Vas deferens B- Seminal vesicle, C- Prostate, D- Bulbourethral gland.
Final Answer : (3)

136) **Solution :** The structures marked in the figure of mammary gland are A–mammary lobe, B–mammary duct, C–ampulla, and D–lactiferous duct. The mammary gland is a gland located in the breasts of females that is responsible for lactation. Mammary glands only produce milk after childbirth. Mammary lobe (1) contains clusters of cells called alveoli which secrete milk which is stored in the cavities of alveoli.
Final Answer : (1)

137) **Solution :** Testicular lobules are the compartments present in the testes, but the lobules as whole are not involved in the process of fertilization. Fusion of male and female gametes are called fertilization.

Final Answer : (3)

138) **Solution :** A – III; B – IV; C – V; D – II; E – I

Column I (Structures of female reproductive system)		Column II (Features)	
A.	Ampulla	III.	Wider part of fallopian tube where fusion of male and female gametes takes place.
B.	Labia majora	IV.	Larger hairy folds which extend down from the mons pubis and surrounds the vaginal opening.
C.	Oviduct	V.	Also called fallopian tubes, which extend from the periphery of each ovary to the womb.
D.	Fimbriae	II.	It helps in collection of ovum after ovulation.
E.	Endometrium	I.	It undergoes cyclical changes during menstrual cycle.

Final Answer : (3)

139) **Solution :** A – I; B – II; C – IV; D – III

	Column I (Phases of menstrual cycle)		Column II (Features)
A.	Menstrual phase	I.	Breakdown of endometrial lining of uterus along with its blood vessels which form liquid that comes out of vagina
B.	Luteal phase	II.	A temporary endocrine structure is formed and secretes a hormone which maintains endometrium and implantation of fertilized ovum and events of pregnancy.
C.	Follicular phase	IV.	Formation of mature Graafian follicle and regeneration of endometrium of uterus.
D.	Ovulatory phase	III.	Secretion of luteinizing hormone at its maximum level and induces breakdown of mature follicle to release the female gamete.

Final Answer : (4)

140) **Solution :** In the given figure of female reproductive system, the marked structures (A to F) are the parts of uterus and fallopian tube. A to F are respectively endometrium, myometrium, perimetrium, isthmus, ampulla and infundibulum.
Final Answer : (3)

141) **Solution :** The mammalian sperm cell consists of a head, neck, middle piece and a tail. The head contains the elongated haploid nucleus with densely coiled chromatin fibres, surrounded anteriorly by an acrosome. Acrosome contains enzymes or penetrating the female egg. Neck contains centrioles (proximal centriole and distal centriole). Distal centriole gives rise to axial filament of the sperm which runs up to the end of the tail. The middle piece has a central filamentous core with many spirally arranged mitochondria which is used for ATP production. The tail or flagellum performs the beating movements that propel the spermatocyte.
Final Answer : (1)

142) **Solution :** Oogonia are called as gamete mother cell. Corpus luteum is formed as a temporary endocrine structure after the ovulation. D is corpus luteum. It is involved in the production of relatively high levels of progesterone and moderate levels of estradiol and inhibin A to maintain pregnancy. A large number of primary follicles degenerates during the phase from birth to puberty. B is tertiary follicle showing antrum.
Final Answer : (3)

143) **Solution :** Spermiogenesis is the final stage of spermatogenesis, which involves the maturation of spermatids and, its transformation into mature, motile spermatozoa.
Final Answer : (4)

144) **Solution :** Gonadotropin-releasing hormone (GnRH) is released from the anterior pituitary. GnRH activity is very low during childhood, and is activated at puberty or adolescence. At the puberty increased secretion of GnRH start the process of sperm formation.
Final Answer : (4)

145) **Solution :** Clitoris is a small, sensitive, erectile part of the female genitals at the anterior end of the vulva. It is homologous with the penis.
Final Answer : (1)

146) **Solution :** Semen, or seminal fluid, is an alkaline fluid that contains spermatozoa embedded in seminal plasma. Semen is ejaculated by male reproductive system during orgasm.
Final Answer : (2)

147) **Solution :** In biology, antrum is a general term for a cavity or chamber, which may have specific meaning in reference to certain organs or sites in the body. Tertiary follicle of ovary contains a fluid filled cavity called antrum and a secondary oocyte ready for ovulation.
Final Answer : (3)

148) **Solution :** The corpus luteum secretes large amounts of progesterone which is essential for maintenance of the endometrium and the pregnancy but its decrease in secretion triggers the menstrual cycle.
Final Answer : (3)

149) **Solution :** Morula is an early stage embryo consisting of cells (called blastomeres) in a solid ball contained within the zona pellucida. The morula is produced by a series of cleavage (mitotic) divisions of the early embryo, starting with the single-celled zygote. Once the embryo has divided into 16 cells, it begins to resemble a mulberry, hence the name.
Final Answer : (2)

150) **Solution :** Apart from being a nourishing medium for the developing foetus, the placenta also acts as an endocrine tissue, producing hormones like human chorionic gonadotropin (hCG), human placental lactogen (hPL), estrogens, progestogens, etc.

Final Answer : (4)

151) **Solution :** Infertility problems cannot be generalised. It is a false statement to say that in India the infertility problems lie in females more often.
Final Answer : (4)

152) **Solution :** Natural method of contraception works on the principle of avoiding chances of ovum and sperm meeting. include periodic abstinence, coitus interrupts and lactational ammenorrhea.
Final Answer : (3)

153) **Solution :** Surgical methods of contraception are generally advised for male/female partner to prevent any more pregnancies. It blocks gametes transport and thereby prevent conception. MTPs are relatively upto first trimester i.e. first three months of the pregnancy / 12 weeks of pregnancy.
Final Answer : (1)

154) **Solution :** The World Health Organization (WHO) is a specialized agency of the United Nations (UN) that is concerned with international public health. It was established on 7 April 1948, headquartered in Geneva, Switzerland.
Final Answer : (3)

155) **Solution :** India is the first country to initiate action plans and programmes at a national level to aware people of the issues of reproductive health and made it one of the social issues to be tackled with good efforts.
Final Answer : (2)

156) **Solution :** The programmes called ‘family planning’ for reproductive health awareness were initiated in 1951.
Final Answer : (1)

157) **Solution :** Reproduction-related areas are currently in operation in India come under the popular name ‘Reproductive and Child Health Care (RCH) programmes.
Final Answer : (4)

158) **Solution :** ‘Saheli’–an oral contraceptive for the females was developed at Central Drug Research Institute (CDRI) in Lucknow, India.
Final Answer : (2)

159) **Solution :** Marriageable age is the age at which a person is allowed by law to marry. In India this age has been set as 18 for woman and 21 for man (without parents’ consent).
Final Answer : (4)

160) **Solution :** The use of condoms during coitus is some of the simple precautions to avoid

contacting STDs like AIDS and syphilis.
Final Answer : (3)

161) **Solution :** Condom, diaphragm, cervical caps, and vaults are included under barrier methods of contraception. Barrier methods of birth control are physical or chemical barriers that prevent sperm from passing through the woman's cervix into the uterus and fallopian tubes to fertilize an egg. Some methods also protect against sexually transmitted disease (STDs).
Final Answer : (3)

162) **Solution :** IUDs stand for Intra Uterine Devices. These are plastic or metal devices placed in the uterus. They contain either copper or a common synthetic hormone used in oral contraceptives. These include loop, copper-T, spiral etc. They prevent fertilization of the ess or implantation of egg.
Final Answer : (1)

163) **Solution :** In artificial insemination (AI) technique, the semen collected either from the husband or a healthy donor is artificially introduced either into the vagina or into the uterus (IUI – intrauterine insemination) of the female.
Final Answer : (2)

164) **Solution :** The hormone releasing Intra Uterine Devices (IUDs) such as Progestasert and LNG-20 increase phagocytosis of sperms within the uterus and makes the uterus unsuitable for implantation & the cervix hostile to the sperms.
Final Answer : (4)

165) **Solution :** Oral administration of small doses of either progestogens or progestogen–estrogen combinations (in the form of pills) is one of the contraceptive methods used by the females.
Final Answer : (1)

166) **Solution :** Female steriliztion (called tubectomy) prevents fertilization by interrupting te passage troug fallopian tube. Eggs continues to be produced but they fail to pass into the uterus. In tubectomy, a small part of the fallopian tube is removed or tied up through a small incision in the abdomen or through vagina.
Final Answer : (4)

167) **Solution :** Medical Termination of Pregnancy (MTP) was legalised in 1971 by Government of India with some strict conditions to avoid its misuse.
Final Answer : (1)

168) **Solution :** MTP is the intentional or voluntary termination of pregnancy before term. MTPs are considered relatively safe during the first trimester, i.e., up to 12 weeks of pregnancy.

Final Answer : (1)

169) **Solution :** Intra cytoplasmic sperm injection (ICSI) is an in-vitro fertilization procedure in which a single sperm is injected directly into an egg. This procedure is most commonly used to overcome male infertility problems, although it may also be used where eggs cannot easily be penetrated by sperm, and occasionally in addition to sperm donation.
Final Answer : (3)

170) **Solution :** ZIFT (Zygote intra fallopian transfer) is the process in which the zygote or early embryos (with upto 8 blastomeres) is transferred into the fallopian tube.
Final Answer : (2)

171) **Solution :** Test tube baby programme employs zygote intra fallopian transfer (ZIFT) technique. In this technique fusion of ovum and sperm is done outside the body of woman to form zygote which is allowed to divide forming 8 blastomeres, then it is transferred into the fallopian tube of the woman.
Final Answer : (4)

172) **Solution :** Mother is homogametic (produce X, X) and father is heterogametic (X and Y). So, father is responsible for the sex of the child not mother.
Final Answer : (4)

173) **Solution :** Cu-7 and Cu-T are intrauterine contraceptive devices for females. They do not suppress sperm motility. Their mode of action is different. Cu-T and Cu-7 discharge 50-75 micrograms of ionic copper into the uterus daily. These copper ions interfere with life sustaining functions that regulate implantation in the uterus. No any hormone is released by them.
Final Answer : (3)

174) **Solution :** Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
Final Answer : (1)

175) **Solution :** A contraceptive pill prevents ovulation, condom prevents sperms to reach the cervix, vasectomy allows the semen to flow but the latter contains no sperms. Copper-T prevents implantation in females.
Final Answer : (2)

176) **Solution :** The given figure is a contraceptive implant. It is a small flexible tube which is inserted under the skin (typically the upper arm) and prevents pregnancy by releasing hormones that prevent ovaries from releasing eggs and by thickening cervical mucus.
Final Answer : (4)

177) **Solution :** The given figure showing the procedure of vasectomy, in which a small part of the vas deferens is removed or tied up through a small incision on the scrotum.

Final Answer : (1)

178) **Solution :** The given figure is that of copper CuT. CuT is a simple copper releasing IUD made of a flexible, "T" shaped piece of plastic wrapped with a thin copper containing wire. It makes the uterus and fallopian tubes produce fluid that kills sperm. This fluid contains white blood cells, copper ions, enzymes, and prostaglandins. Copper ions prevent pregnancy by inhibiting the movement of sperm, because the copper-ion-containing fluids are directly toxic to sperm. Even if an aggressive little spermatozoon fertilizes an egg, the copper ion laden environment prevents implantation of the fertilized egg, and thus pregnancy.
Final Answer : (1)

179) **Solution :** ovulation is expected on 10th - 17th day of menstrual cycle.
Final Answer : (3)

180) **Solution :** The success of birth control programmes in controlling population growth is dependent on acceptability of the above by the people.
Final Answer : (4)